Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

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Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
2D	two-dimensional
AICc	Akaike information criterion with the correction for small samples
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CHO	Chinese hamster ovary
CI	confidence interval
CPP	critical process parameter
CQA	critical quality attribute
CV	column volume (as a unit) or coefficient of variation (as %CV)
DNA	deoxyribonucleic acid
ELISA	enzyme-linked immunosorbent assay
Fab	antigen-binding fragment
Fc	crystallizable fragment
GMP	good manufacturing practice
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
HPLC	high performance liquid chromatography
icIEF	imaged capillary isoelectric focusing
KPP	key process parameter
LMW	low molecular weight species
ln	natural logarithm
LoF	lack of fit (in table headers)
LOQ	limit of quantitation
LPA	leached Protein A (in table headers)
Mfg	manufacturing scale (in table headers)
MSAT	Manufacturing Science and Technology
n.a.	not applicable

Abbreviation	Meaning
n.m.	not meaningful
n.t.	not tested
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
PRESS	prediction error sum of squares
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
RMSE	root mean square error
RSM	response surface methodology
SDM	scale-down model (in table headers)
SEC	size exclusion chromatography
SOP	standard operating procedure
VH3	family 3 of the human heavy chain variable domains
WC-CPP	well-controlled critical process parameter

Executive summary

This report presents the process characterization of the Protein A chromatography step (Step 5), the capture step of the A-Mab drug substance purification train. It removes most of the host cell protein and DNA of the clarified harvest, and it adds one impurity of its own, Protein A ligand that leaches from the resin. The characterization studies were run under the approved plan PCP-005 in a scale-down model of the commercial column, between 2025-11-05 and 2026-04-17.

This Version 2.0 re-analyses the experiments of Version 1.0 against the criteria of the master plan PCMP-001 Version 2.0, issued under the investigation INV-001. It adds the fragment results by nrCE-SDS and analyses each response-surface output by model selection, as the investigation INV-002 requires. The normal operating ranges (NORs) of the step are not changed.

The step has 6 process parameters. Of these, 4 were studied in a 19-run screening design followed by a 28-run response-surface design, and 2 were studied one factor at a time. Protein load and elution buffer pH were classified as well-controlled critical process parameters (WC-CPPs). Load flow rate and end of pool collect were classified as KPPs, and operating temperature and bed height as GPPs. No parameter of the step is in the class CPP.

- F1. The scale-down model is qualified for leached Protein A. Pool host cell protein in the scale-down model is 0.973 times its value at manufacturing scale (90 % confidence interval 0.891 to 1.063). Over the NOR in force, with protein load at 20–40 g/L resin, the model is not qualified for pool host cell protein. With the ratio at its estimate the upper prediction limit of a single run is 15,697 ng/mg, below the criterion of ≤ 16,719 ng/mg. With the ratio at the low end of its interval it is 17,143 ng/mg, above the criterion. The model is used for pool host cell protein all the same, with the ratio at its estimate (PD-005-10). This holds with every model that the data support about as well as the selected one, on the planned runs alone and against the a priori criterion. The NOR is not narrowed, because NORs come from production. The report flags the protein load NOR for discussion with manufacturing on whether it can be improved. With the ratio at the low end of its interval the limit meets the criterion when the upper edge of the protein load NOR is 38.9 g/L resin. Leached Protein A is 1.18 ppm lower in the scale-down model, as expected from the longer hold of the load at manufacturing scale. Fragments were not compared at the two scales (Section 3.1).
- F2. Each output of the response-surface design was analysed by model selection (PD-005-04), and each selected model meets the adequacy criteria of PCP-005 §7 on lack of fit and predictive power. Pool host cell protein is modelled on the log scale (Section 3.5, Section 5.3).

- F3. Pool host cell protein rises with protein load and as elution buffer pH falls, by factors of 2.43 and 3.30 across the ranges studied. The two effects multiply. Whether a further interaction of the two exists on the log scale the data do not decide. Pool host cell protein falls by 15 % as the end of pool collect is extended across its range (Section 5.3.1).
- F4. Step yield rises with end of pool collect and levels off towards the upper end of its range. It falls with protein load, and load flow rate lowers it only at high load (Section 5.3.2).
- F5. Leached Protein A does not depend on the 4 parameters of the multivariate design by more than the design resolves. This rests on the re-assay of run 11 (DEV-005-02). Every run met the criterion of ≤ 158 ppm at manufacturing scale, the highest at 5.79 ppm (Section 5.3.3).
- F6. Fragments by nrCE-SDS fall slightly as elution buffer pH rises, by 0.11 percentage points across its range studied. All 28 pools of the response-surface design met the criterion of ≤ 2.45 %, the highest at 1.09 % (Section 5.3.4).
- F7. Each of the 3 outputs with a criterion met it across the NOR in force, pool host cell protein with the offset ratio at its estimate (F1). The criterion for pool host cell protein, ≤ 16,719 ng/mg, is 4 % wider than its a priori value, because the one-sided 95 % prediction bound of a single batch for the host cell protein clearance of anion exchange is better than its platform worst case. The change is the producer's risk, and an excursion would be detected at release of the drug substance (Section 7). Every NOR-propagated PAR spans the range studied of its parameter, except that of elution buffer pH for pool host cell protein, 3.22–3.9. On the upper prediction limit of a single run, pool host cell protein exceeds its criterion only at high load combined with low pH, never with every parameter inside its NOR (Section 6).
- F8. Criterion 5 of PCP-005 §7, on residual trends and influential runs, is met except for leached Protein A and fragments, whose residual trends are small against their criteria (PD-005-02). With each run above 4/n deleted in turn the conclusion over the NOR is the same for every output with a criterion (Section 5.5).
- F9. In a hold study made during the Version 2.0 re-analysis (PD-005-05), one eluate made at set-point was held at elution pH for 7 days at 2-8 °C or 21 °C. What the hold added kept aggregate, acidic variants and fragments within the upper limits of the low-pH inactivation load. Acidic variants can enter the hold below their lower limit of 18 % at the least favourable corner of the production bioreactor NOR, a result of PCR-003, and the hold does not lower them (Section 5.7).
- F10. No difference from the set-point runs was detected in pool host cell protein, step yield or leached Protein A at either end of the ranges of operating temperature and bed height, and each of these results met its criterion. Fragments were not measured by nrCE-SDS in this study (Section 5.6).
- F11. The effect of end of pool collect on pool host cell protein has opposite signs in the screening and the response-surface designs. The difference is larger than the noise of the screening design, and its cause was not identified (Section 5.3.1).

At commercial scale the integrated process model gives a Cpk of 6.14 for host cell protein and 10.62 for fragments in the drug substance, above the target of 1.33 of

PCMP-001 §7.9. No simulated batch of 2,000 fell outside the drug substance criterion for host cell protein, leached Protein A, fragments or residual DNA (Section 8).

The 2 study events of the step (DEV-005-01 and DEV-005-02) change no judgement of an output against its criterion. The report also records 10 discrepancies from the plan (Section 13), and its results roll up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cell-mediated cytotoxicity. It is developed by Novacyte Biologics and is being transferred from development at the sending site to commercial drug substance manufacture at 15,000 L scale at the receiving site, under the Process Transfer Plan PTP-001. The drug substance is purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and UF/DF.

Protein A chromatography is the first chromatographic step of the purification train. It is operated in bind and elute mode. The resin binds the antibody from the clarified harvest at neutral pH, while host cell protein, DNA and most other components of the feed are removed in the load flow-through and the wash. A low-pH buffer elutes the antibody into a small volume, and the eluate passes to low-pH viral inactivation (Step 6) without neutralization. The step is the principal point of clearance for host cell protein and DNA, and it partly clears fragments. It introduces one attribute of the CQA register, leached Protein A. Although Protein A chromatography removes some virus, no viral clearance claim is made for this step (CMC Biotech Working Group 2009). The viral clearance claim of the process rests on low-pH inactivation, anion exchange and virus filtration, and it is consolidated in PCMR-001.

1.2 Objectives

The objectives of the characterization were the following, as PCP-005 §2 states them.

1. Quantify the effect of each characterized parameter on pool host cell protein, on step yield and on leached Protein A, over the ranges given in PCP-005 §6.1.
2. Define the region of the characterized space within which the quality attributes this step controls are predicted to remain within their acceptance criteria.
3. Establish a proven acceptable range for each parameter and each quality attribute the step controls, both with the other parameters at their set-points and with the other parameters varying within their normal operating ranges.
4. Provide the demonstrated effects and the operating region from which each parameter will be classified under SOP-4001.
5. Justify the parameter ranges carried into Stage 2 and the contribution this step makes to the control strategy of the drug substance process.

6. Estimate, before the first characterization run, the offset between the scale-down model and manufacturing scale for each quantity that the qualification compares, and qualify the model by that offset for pool host cell protein and leached Protein A (PCP-005 §5.1).
7. Fit response-surface models for pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS over the ranges studied, and judge their adequacy for prediction.

Objectives 1 to 5 are those of Version 1.0. Objectives 6 and 7 were added at Version 2.0, which also adds fragments to objectives 1 and 3 and states the ranges of objective 3 at manufacturing scale for pool host cell protein and leached Protein A. Objective 1 is answered in Section 5.3 and Section 5.6, objective 2 in Section 6, objective 3 in Section 7, objective 4 in Section 9, objective 5 in Section 7 and Section 10, objective 6 in Section 3.1 and objective 7 in Section 5.3.

The hold study of the eluate was not an objective of PCP-005. It was made during the Version 2.0 re-analysis and is recorded as the discrepancy PD-005-05 (Section 13.3.2).

1.3 Regulatory and scientific basis

The study is part of Stage 1 (process design) of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11 (International Council for Harmonisation 2009, 2012), in which prior knowledge and risk assessment identify the parameters that may affect quality, and designed experiments then measure and bound those effects. Criticality is treated as a continuum, as ICH Q8(R2) describes it, and risk is managed under ICH Q9 (International Council for Harmonisation 2023). The classification of parameters follows the decision logic of the A-Mab case study (CMC Biotech Working Group 2009), as PCMP-001 §7.7 applies it. Table 1.1 lists the documents this report depends on and feeds.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-005	Process Characterization Plan (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 takes effect on 2026-09-30 and is issued under the investigation INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the PAR analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). For this step, Version 1.0 judged pool host cell protein against ≤ 10,724 ng/mg. That limit was the drug substance criterion carried back through the nominal downstream clearance of this campaign, then divided by a safety factor of 4.5. The factor was chosen. It was not calculated from how the downstream steps perform (INV-001 finding 2). Leached Protein A was judged against the drug substance criterion, and fragments were not reported. Table 1.2 sets the Version 1.0 limits beside the Version 2.0 criteria.

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, Protein A.

Attribute	Version 1.0[a)]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Leached Protein A (ppm)	none[b)]	–	≤ 158	Carried back
Fragments, LMW (%)	not assessed[c)]	–	≤ 2.45	Carried back
Host cell protein (ng/mg)	≤ 10,724	R1, margin 4.5	≤ 16,719	Carried back, widened[d)] from ≤ 16,000

[a)] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b)] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[c)] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[d)] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[R1] Version 1.0 rule: nominal back-calculation / margin.

The Version 2.0 criteria are intermediate acceptance criteria. PCMP-001 §7.3 derives them a priori, which here means from the drug substance criteria and the platform worst case of each downstream step, without data from this campaign. PCMP-001 §7.5 then lets the one-sided 95 % prediction bound of a single batch of a downstream step tighten or widen them (Section 7). PCMP-001 Version 2.0, which states both rules, took effect on 2026-09-30, after every study of this report had ended. The criterion in force for pool host cell protein, ≤ 16,719 ng/mg, is wider than its a priori value of ≤ 16,000 ng/mg, because the A-Mab results of anion exchange widened it. It is also wider than the Version 1.0 limit of ≤ 10,724 ng/mg, which was set at analysis and was never an approved criterion (INV-001 finding 1).

The designs, their runs and their results are those of Version 1.0. The changes to this report are listed below.

1. Model selection. Each output of the response-surface design is analysed by model selection, the corrective action CA3 that INV-002 adopted on 2026-09-15 (discrepancy PD-005-04, Section 3.5). Version 1.0 fitted the full quadratic model on the raw scale to every output. Pool host cell protein is now modelled on the log scale.
2. Response-surface run 23. The run is analysed at the elution buffer pH recorded under DEV-005-01, 3.38, and not at its design setting of 3.55 (discrepancy PD-005-03, Section 13.1).
3. The NOR-propagated PAR for pool host cell protein. Under the Version 1.0 limit it was narrower than the range studied for every parameter of the multivariate design. For protein load it was 10–34.5 g/L resin, which ended inside the NOR of 20–40 g/L resin and made the upper end of that NOR appear unsupported. For elution buffer pH it was 3.48–3.9, for load flow rate 138–300 cm/hr and for end of pool collect 2.29–3.2 CV. Under Version 2.0 every NOR-propagated PAR

spans its range studied, except that of elution buffer pH for pool host cell protein, 3.22–3.9 (Section 7).

4. Leached Protein A. Judged against the drug substance criterion of ≤ 5 ppm, its NOR-propagated PAR of elution buffer pH was 3.2–3.86. Leached Protein A is now judged against its intermediate acceptance criterion, and its PAR spans the range studied of every parameter.
5. Fragments by nrCE-SDS are reported for the first time, from the retained response-surface pools, tested in study S-PA-RET under the method decision AMD-001 (Table 3.3). The SEC fragment results are kept in the record and are not used for any decision, because SEC reads part of the host cell protein in the pool as fragment (INV-001 finding 6).
6. The scale-down model is qualified by the offset of each output from manufacturing scale (PCMP-001 §7.1), and every result of pool host cell protein and leached Protein A judged against a criterion is stated at manufacturing scale. A further 6 set-point runs of the scale-down model (study S-PA-SDM2) are recorded as the discrepancy PD-005-01 (Section 3.1).
7. Over the NOR of protein load, 20–40 g/L resin, the scale-down model is not qualified for pool host cell protein at the criterion in force. The NOR is not narrowed, because NORs come from production, and the report flags it for discussion with manufacturing on whether it can be improved (Section 3.1). The model of pool host cell protein is used with the ratio at its estimate, against the rule of PCMP-001 §7.1, and this is recorded as the discrepancy PD-005-10 (Section 13.3.6). No NOR and no set-point of the step is changed.
8. A hold study of the Protein A eluate (S-PA-HLD) was made during the Version 2.0 re-analysis and is recorded as the discrepancy PD-005-05 (Section 13.3.2). Version 1.0 had no hold result. Acidic variants in the held eluate are judged against both limits of their two-sided criterion (INV-001 finding 4).
9. The execution window of every study is recorded (Table 4.2), because the Version 1.0 records gave no execution dates (INV-001 finding 5).
10. The events that Version 1.0 numbered as deviations, DEV-005-01 and DEV-005-02, are recorded as study events under their numbers, because a process characterization study is not GMP manufacture. The departures from the plan found in the re-analysis are recorded as discrepancies from the plan (Section 13).

The directions of the main effects are those of Version 1.0. The interaction of protein load with elution buffer pH that the Version 1.0 model carried for pool host cell protein ($p < 0.001$) is largely the multiplicative structure of the two main effects on the log scale (Section 5.3.1).

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The same capture step has been used to purify the platform antibodies X-Mab, Y-Mab and Z-Mab, and A-Mab through clinical manufacture (CMC Biotech Working Group 2009). Staphylococcal Protein A, immobilized on the resin, binds the Fc region of the antibody at the interface between the second and third constant domains of the heavy chain. When the pH is lowered, histidine residues at this interface become protonated, the interaction weakens, and the antibody is released.

Host cell protein reaches the eluate by two routes. Some binds weakly to the resin backbone or to the ligand. Some associates with the antibody through electrostatic and hydrophobic interactions, which the wash does not disrupt while the antibody stays bound to the ligand. This product-associated host cell protein leaves the column with the antibody at any elution pH. A lower elution pH desorbs more of the host cell protein held on the resin, and that protein leaves the column with the product. Pool host cell protein was therefore expected to rise as elution buffer pH falls. In the A-Mab case study pool host cell protein also rose with protein load (CMC Biotech Working Group 2009), and the same direction was expected here. Product-associated host cell protein rises in proportion to the antibody, which leaves the value per mg unchanged. PCP-005 therefore assumed no mechanism for the rise with load (PCP-005 §4.1). The plan expected that the two effects could interact, and both designs can estimate an interaction.

Step yield is lost when antibody breaks through during the load or remains on the column at the end of collection. Breakthrough begins earlier as the load approaches the dynamic binding capacity of the bed. The dynamic binding capacity itself falls as the residence time shortens, because binding is limited by the diffusion of the antibody into the pores of the bead. A higher load flow rate shortens the residence time, which costs yield only when the load is close to the capacity. Yield was therefore expected to fall with flow rate mainly at high load, and to rise as the end of pool collect is extended into the tail of the elution peak.

Leached Protein A comes from hydrolysis of the coupling chemistry and from cleavage of exposed ligand by proteases that the clarified harvest carries into the column. The leached fragments elute with the antibody because they still bind Fc. Leaching is highest on new resin, in its first cycles. A higher protein load or a lower load flow rate lengthens the loading time, and with it the contact of the ligand with the harvest proteases. Leached Protein A is reported per mass of antibody, and if a similar amount of ligand leaches in each cycle the value in ppm falls as protein load rises. PCP-005

expected leached Protein A to depend little on the parameters within their ranges (PCP-005 §4.1).

An antibody fragment that still carries Fc is captured and elutes with the intact antibody. A Fab fragment lacks Fc and is expected to flow through with the load, so the pool is expected to carry less fragment than the feed. Native Protein A also binds the variable domain of heavy chains of the VH3 family, and a Fab fragment of such an antibody can be captured as well. Whether the Fab of A-Mab binds the ligand of this resin was not tested here. A lower elution pH may add a little acid-catalysed hinge hydrolysis during elution. SEC overstates fragments in this matrix, because it reads part of the host cell protein as fragment (INV-001 finding 6). DNA is polyanionic and does not bind Protein A. Its clearance is large and was not expected to depend on the operating parameters within their ranges.

Protein A, when operated under platform conditions, does not separate glycan variants or charge variants, because it binds the Fc region independently of both (CMC Biotech Working Group 2009). Nor does it separate aggregate from monomer, since both carry Fc. In the Protein A characterization of the case study no process parameter had a significant effect on aggregate (CMC Biotech Working Group 2009). For these reasons glycan variants, charge variants and aggregate were not responses of the designs.

2.2 Quality attributes in scope

Table 2.1 lists the attributes in scope of this report with their drug substance criteria and the criticality level the register assigns. Each attribute is a CQA with a level from very low (VL) to very high (VH), assigned with Tool #1, whose score is the product of impact and uncertainty. Acidic variants are in scope because the eluate is held before viral inactivation (Section 5.7).

Table 2.1: Quality attributes in scope of this report, with the drug substance criterion and criticality.

CQA	Category	Drug substance criterion	Criticality	Tool #1 score
Leached Protein A	process impurity	≤ 5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36
Fragments (LMW, clipped species)	size variant	≤ 2 % (nrCE-SDS)	L	12
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

The drug substance criteria do not apply to the Protein A pool directly, because cation exchange and anion exchange remove host cell protein, leached Protein A and fragments after this step. Against the drug substance criteria at manufacturing scale, pool host cell protein would fail in all 47 results of the designs, leached Protein A in 4 of 47, the lowest of them at 5.002 ppm, and fragments in none of 28. Each of these outputs is therefore judged against an intermediate acceptance criterion (Section 7). Residual DNA has a drug substance criterion but no intermediate acceptance criterion, because the platform history gives no worst-case DNA clearance for the steps after Protein A (PCMP-001 §7.3). It was measured in pg per mg of antibody on the centre points, for information, and it is judged in ng per dose in the drug substance (Section 8). The held eluate is judged against the criteria of the low-pH inactivation load (Section 4.5). Step yield is a performance attribute without an acceptance criterion, and it is used for the classification of parameters that act on process performance.

2.3 Risk-based prioritization and parameter selection

The pre-characterization risk assessment RA-001 assessed every parameter of the step before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters.

Protein load, elution buffer pH, load flow rate and end of pool collect were assigned to multivariate study. Each carries an effect expected from the mechanism, and the interactions of protein load with elution buffer pH and with load flow rate can only be estimated in a design that varies the parameters together. Operating temperature and bed height were assigned to univariate study. A higher temperature speeds the diffusion of the antibody into the pores, which raises the dynamic binding capacity, and it strengthens the hydrophobic contacts at the Fc–ligand interface, which makes elution slightly less complete. Over the range studied both effects were expected to be small against those of load and pH, and the resin manufacturer specifies binding capacity across that range. Bed height is held by the packing specification in manufacture, and at constant linear velocity it sets the residence time.

The ranges studied are between 1.9 and 2.3 times the width of the NOR. They bound the knowledge space within which a PAR can be claimed. Load flow rate is set as a linear velocity, so at the set-point bed height it fixes the residence time. The residence time is 6 min at the set-point flow rate, and over the range studied it runs from 4 min at the highest flow to 12 min at the lowest. The upper end of the flow range therefore tests the shortest residence time, the condition most likely to cause breakthrough at high load, and the two ends of the elution pH range test the strongest and the weakest elution.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model is a laboratory column packed with the commercial Protein A resin and operated under SOP-2007 on the scaling principles of PCMP-001 §5. The bed height and the residence time of the commercial column were kept. Load, wash and elution volumes were normalized to column volumes, and buffers were made to the commercial recipes. Column efficiency was checked by plate count and peak asymmetry after packing. PCP-005 sets no criterion for either, and both are reported for information.

The model was qualified under SOP-1001 by its offset from manufacturing scale, the rule of PCMP-001 §7.1 that PCP-005 §5.1 applies to this step. Set-point runs of the scale-down model were compared with 5 manufacturing-scale batches of A-Mab run at set-point. The outputs compared were pool host cell protein, leached Protein A, step yield, and column efficiency as plates per metre and peak asymmetry. The offset of an output is the mean of the scale-down runs minus the mean of the manufacturing-scale batches, with its two-sided 90 % confidence interval from a Student t interval with pooled variance. It is taken on the scale on which the output is modelled (Section 3.5). Pool host cell protein is modelled on the log scale. Its offset is therefore the difference of the mean logarithms, reported as the ratio of the geometric means of the two scales, a multiplicative offset. A scale-down result of pool host cell protein is divided by this ratio before it is judged against a criterion, and a result of leached Protein A has its offset subtracted. Every range and every conclusion for these two outputs is therefore at manufacturing scale.

The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of its confidence interval. That conclusion is whether the output meets its criterion at every combination of NOR low, set-point and NOR high of the 4 response-surface factors, judged on the upper prediction limit of a single run (Section 3.5). Step yield and column efficiency have no criterion at the step. Their offsets are recorded, and no conclusion depends on them.

Table 3.1 gives the offsets over all 9 scale-down runs, with 12 degrees of freedom. Of these, 3 were planned in study S-PA-SDM (2025-11-05 to 2025-11-14), before the first run of the screening design, and 6 were made in study S-PA-SDM2 (2026-09-07 to 2026-09-11) during the Version 2.0 re-analysis (discrepancy PD-005-01, Section 13.3.1).

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, Protein A.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Host cell protein (ng/mg)	9	5	7,701[a)]	7,911[a)]	×0.973[a)]	×0.891 to ×1.063	No / Yes / Yes	No
Leached Protein A (ppm)			2.52	3.70	−1.18	−1.97 to −0.39	Yes / Yes / Yes	Yes
Step yield (%)			94.2	94.4	−0.2	−2.1 to 1.7	no criterion[b)]	n.a.[b)]
Column efficiency (plates/m)			2,533	2,649	−116	−320 to 89	no criterion[b)]	n.a.[b)]
Peak asymmetry			1.20	1.22	−0.02	−0.09 to 0.05	no criterion[b)]	n.a.[b)]

[a)] Modelled on the log scale: geometric means, and the offset as the ratio of the scale-down to the manufacturing geometric mean. The manufacturing-scale value is the scale-down value divided by the ratio.
[b)] The output has no acceptance criterion at this step, so no conclusion depends on its offset and qualification does not apply. The offset is recorded.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

Pool host cell protein in the scale-down model is 0.973 times its value at manufacturing scale. The 90 % confidence interval of this ratio, 0.891 to 1.063, includes 1. Dividing by the low end of the interval raises each prediction most, so the low end is the unfavourable end for an upper criterion.

Over the NOR in force, with protein load at 20–40 g/L resin, the scale-down model is not qualified for pool host cell protein. The upper prediction limit of a single run over the NOR is 15,697 ng/mg at manufacturing scale with the ratio at its estimate and 14,374 ng/mg with the ratio at its high end, both below the criterion of ≤ 16,719 ng/mg. With the ratio at its low end it is 17,143 ng/mg, above the criterion. The conclusion therefore changes across the interval of the ratio. With the other factors over their NORs, the limit at the low end of the ratio meets the criterion when the upper edge of the protein load NOR is 38.9 g/L resin.

The NOR is not narrowed. NORs come from production. An offset compares laboratory results with production scale, so it can change a PAR but not a NOR. No NOR of the step lies outside its NOR-propagated PAR at manufacturing scale (Section 7). The flag of this report therefore comes from the qualification of the scale-down model and not from a PAR. The report flags the protein load NOR for discussion with manufacturing on whether it can be improved. The load of 38.9 g/L resin is the information that discussion needs.

PCMP-001 §5 asks the report to state the consequence. Every judgement of pool host cell protein in this report is made with the ratio at its estimate, as PCMP-001 §7.1 rule 2 requires, and on that basis pool host cell protein meets its criterion across the NOR in force. The study does not show that this holds over the whole interval of the ratio. At the upper end of the protein load NOR, a true ratio near the low end of its

interval would put the upper prediction limit of a single run above the criterion. The judgements over the NOR, the design space and the PARs for pool host cell protein carry this limitation. PCMP-001 §7.1 states that “a model whose conclusion for an output changes between the two ends of the offset interval is not used for that output”. This report uses the model of pool host cell protein all the same, and that departure is the discrepancy PD-005-10 (Section 13.3.6).

The conclusion does not depend on the choices behind the analysis.

1. The criterion. Without the model selection of INV-002 corrective action CA3 the criterion would be its a priori value of ≤ 16,000 ng/mg (Section 7). Against it the model is not qualified either. The limit with the ratio at its estimate, 15,697 ng/mg, is below that criterion, and the limit with the ratio at its low end, 17,143 ng/mg, is above it.
2. The added runs. On the 3 planned runs alone the interval of the ratio is wider, and the model is not qualified either (Section 13.3.1).
3. The model. Each of the 9 models whose AICc lies within 2 units of that of the selected model gives a limit above the criterion with the ratio at its low end (Section 5.3.1).

Leached Protein A is 1.18 ppm lower in the scale-down model. The 90 % confidence interval of the difference, 0.39 to 1.97 ppm lower, excludes zero, and each scale-down run was below the manufacturing mean. The expected cause is cleavage of the ligand by harvest proteases. At manufacturing scale the clarified harvest is held longer before capture. Proteolytic activity in the harvest is expected to rise while it is held, and the load then cleaves more ligand from the resin during loading. This study does not test that cause. With the offset at either end of its interval the conclusion for leached Protein A is the same, and the model is qualified for it. On the 3 runs of the planned study S-PA-SDM alone, the two scales differ in leached Protein A ($p = 0.037$). PCP-005 Version 1.0 required such a difference to be investigated before the characterization studies, and it was not (discrepancy PD-005-08, Section 13.3.4).

Fragments by nrCE-SDS were not compared at the two scales. The method decision AMD-001 of 2026-03-20, which set nrCE-SDS as the method for fragments at every step, was taken after study S-PA-SDM had closed. The 6 runs of study S-PA-SDM2 were made after AMD-001 and were not tested by nrCE-SDS either. Fragments therefore have no measured offset (PCMP-001 §7.1), and their conclusions assume no difference between the scales. The scale-down model is accepted for fragments on mechanism, as PCP-005 §5.1 states. The capture of Fc-bearing species depends on bed height and residence time, and low-pH hinge hydrolysis on pH, time and temperature, all of which the model keeps at commercial values.

The longer hold of the load at manufacturing scale could also raise fragments, because the harvest proteases that cleave the ligand can also clip the antibody. The hold study of the clarified harvest (PCR-004) bounds this effect. Over 7 days fragments by nrCE-SDS in the clarified harvest rose by at most 0.55 percentage points, at 21 °C. Suppose the longer hold added that much fragment to the pool, every fragment it formed being captured. The upper prediction limit of fragments over the NOR, 1.09 %, would then become 1.65 %, still below the criterion of ≤ 2.45 %. This bound holds for a hold of the clarified harvest within the 7 days of that study. The hold study of the eluate

measured attributes that the qualification did not compare, so its results also carry no offset.

3.2 Operation

Each run followed SOP-2007 at laboratory scale. The column was equilibrated, loaded with clarified harvest to the target protein load, washed, and eluted with the elution buffer at the pH set by the design. The protein load of each run was set from the load volume and the titre of the clarified harvest by Protein A affinity HPLC (AMV-3021, under SOP-3016). The eluate was collected from the rise of the absorbance at 280 nm until the end of pool collect set by the design, in column volumes. The column was then stripped, cleaned and stored under SOP-2008. The pool was not neutralized, because the commercial eluate passes to low-pH viral inactivation at elution pH.

Each design was loaded from one pool of clarified harvest produced at set-point and held frozen in aliquots. A single feed keeps feed variation out of the estimated effects. The effect of feed quality on the step was therefore not estimated in these studies. The case study reports that Protein A capture is robust over a wide range of feed inputs (CMC Biotech Working Group 2009), and the feed is characterized in PCR-003 and PCR-004. Runs were executed in random order within each design. The resin was used within the life-cycle limits of SOP-2008, and the life cycle of the resin is outside the scope of this report.

3.3 Analytical methods

Each response was measured by a validated method under its own AMV. Table 3.2 lists the controlled procedures and method validations used. The SOP and AMV numbers are placeholders.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2007	Operation of the Protein A Capture Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-3015	Purity by Non-Reduced CE-SDS
	SOP-3016	Product Titre by Protein A Affinity HPLC
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3013	Charge Variants (icIEF)
	AMV-3014	Residual DNA (qPCR)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3021	Product Titre (Protein A Affinity HPLC)

Pool host cell protein was measured by ELISA (AMV-3012) and is reported per mass of antibody in the pool, so that it does not depend on how far the pool is diluted by the end of pool collect. The precision of the ELISA is 9.5 % of the level. The titre of the clarified harvest was measured by Protein A affinity HPLC (AMV-3021, under SOP-3016), which measures the antibody alone, with a precision of 2 %. Step yield is the mass of antibody in the pool, from the protein concentration by absorbance at 280 nm (AMV-3019) and the pool volume, divided by the mass loaded, from the titre and the load volume. PCP-005 §5.3 planned both masses by absorbance at 280 nm, so the titre basis of the loaded mass is the discrepancy PD-005-06. In the pool, host cell protein is at most 2.5 % of the mass of antibody, so the absorbance at 280 nm there is mainly that of the antibody. Leached Protein A was measured by ELISA (AMV-3016). Fragments were measured by nrCE-SDS (AMV-3020, under SOP-3015), as PCMP-001 §7.8 sets for every step. SEC (AMV-3011) remains the method for aggregate (PCMP-001 §7.8). Acidic variants in the hold study were measured by icIEF (AMV-3013), and residual DNA by qPCR (AMV-3014). Table 3.3 shows which method measured fragments in each study of the step, and Appendix D summarizes the methods.

Table 3.3: Methods used for size variants in the designed studies, Protein A.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	SEC
Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]

[a)] In the Protein A and low-pH viral inactivation pools part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained samples of the response-surface design, tested 2026-04-06 to 2026-04-17.

The screening design ran before the method decision AMD-001 set nrCE-SDS as the method for fragments, and its pools were not retained. Its fragment results are by SEC only and are not used for any decision. The univariate study S-PA-UNI ran before AMD-001, and its samples were not retained, so its fragment results are by SEC only and are not used for any decision. The response-surface pools were retained frozen and were tested by nrCE-SDS in study S-PA-RET (2026-04-06 to 2026-04-17). On those pools SEC read fragments higher than nrCE-SDS by 0.12 to 1.37 percentage points (median 0.36). The hold study ran after AMD-001 and measured fragments by nrCE-SDS alone.

3.4 Sampling plan

The load of each design was sampled once per feed aliquot thawed. The pool of every run was sampled once at the end of collection, after mixing, and its volume was recorded. Each pool sample was divided into aliquots for each method. One further aliquot of every response-surface pool was retained frozen under the sample management procedure of Analytical Development, and these are the samples tested by nrCE-SDS in study S-PA-RET. The screening design had 3 centre points and the response-surface design 4. Residual DNA was measured on the load and pool of the centre-point runs. The hold study was sampled as described in Section 4.5.

3.5 Statistical methods

The designs were built and analysed under SOP-1002 with the common approach of PCMP-001 §6. Factors were coded from −1 to +1 over their range studied, with 0 at the centre. For each of the 4 design factors the centre of the range studied is the set-point, so the centre-point runs of both designs are set-point runs. In this report a term of a model is a main effect, an interaction or a square, and the intercept is counted apart.

The screening data were fitted with a model of main effects and two-factor interactions. Effects are reported as the change in the response over the coded range of

the factor, which is twice the coefficient. Significance was judged at α = 0.05. The screening model identifies effects. A two-level design cannot resolve curvature, so the screening model was not used to predict between its corners.

PCP-005 §5.5 planned to fit the full quadratic model, an intercept and 14 terms, to each output of the response-surface design. Each output was analysed instead by model selection, which INV-002 introduced as its corrective action CA3 on 2026-09-15 for every report with a response-surface design. PCMP-001 Version 2.0 still names the full quadratic model, so the selection is a discrepancy from the plan, recorded as PD-005-04 (Section 13). The selection first chooses the scale of each output and then the terms of its model. Pool host cell protein is the one output put on the log scale, and both choices are explained below with its data.

3.5.1 The scale of the model

On the raw scale a model describes pool host cell protein in ng/mg as a sum, $y = b_0 + b_1 x_1 + \cdots + \varepsilon$, where x_1 is a coded parameter, b_1 its coefficient and ε the run-to-run error. Each parameter then adds or removes a fixed amount, and the error is a fixed amount too. On the log scale the model is $\ln y = b_0 + b_1 x_1 + \cdots + \varepsilon$. Transformed back, $y = e^{b_0} \cdot e^{b_1 x_1} \cdots e^{\varepsilon}$, so each parameter multiplies pool host cell protein by a fixed factor. The factor is e^{b} per coded unit and e^{2b} from the low to the high end of the range studied. The run-to-run error multiplies the output as well, so its size in ng/mg grows with the level. In the selected model pool host cell protein rises by a factor of 2.43 from the low to the high end of protein load, falls by a factor of 3.30 from the low to the high end of elution buffer pH, and falls to 0.85 times its value over the range of end of pool collect.

Two properties of this output make a factor the natural description. A purification step removes a share of what is present, so a parameter that changes the performance of the step changes the share of host cell protein that reaches the pool, not a fixed amount. The precision of the host cell protein ELISA is a percentage of the level, 9.5 % (AMV-3012), so the scatter of the results grows with their level.

Least squares, and the t-tests, F-tests and prediction intervals built on it, assume that the residuals are independent and normally distributed, with mean zero and the same variance over the whole range studied. On the raw scale the residuals of pool host cell protein do not have the same variance. In the full quadratic model on the raw scale, their root mean square is 679 ng/mg in the lower half of the predicted values and 957 ng/mg in the upper half, 41 % larger. In the full quadratic model on the log scale the two halves give 0.067 and 0.074 in natural-log units, 10 % apart, so the variance is about constant. The Box-Cox profile of the full quadratic model (Box and Cox 1964) points the same way. Its estimate of the exponent λ is 0.06, and its 95 % interval, −0.32 to 0.42, contains 0, the log scale, and excludes 1, the raw scale. The rule of the selection puts an output on the log scale exactly when this interval contains 0 and excludes 1. For step yield, leached Protein A and fragments the interval contains 1, and these outputs stay on the raw scale. The residual plot of the selected model shows no funnel (Figure 5.2).

The log scale changes how results are read, not what is judged. The criterion stays in ng/mg. Effects are fold changes. The scale-down offset of pool host cell protein is a ratio of geometric means, and a scale-down prediction is divided by it (Section 3.1). A prediction interval is calculated on the log scale and transformed back. It keeps its 95 % coverage, because the probability that $\ln y$ lies between two values is the probability that y lies between their antilogarithms. It is therefore suitable, and on the log scale the assumption of least squares holds, which on the raw scale it does not. The interval is asymmetric, further above the predicted value than below it. At the set-point the predicted value is 7,150 ng/mg at manufacturing scale, and the 95 % prediction interval of a single run runs from 5,687 to 8,989 ng/mg, 1,463 below and 1,839 above. The predicted value is the median of pool host cell protein, its geometric mean, not its arithmetic mean. The upper prediction limit, which decides against an upper criterion, does not depend on that distinction. Near an upper criterion, where the level is high, a model on the raw scale understates the scatter, because its residual standard deviation is pooled over all levels. The interval from the log scale is wider there, as the data require.

3.5.2 The terms of the model

The terms were then chosen by a search with one goal, the model with the lowest AICc among all hierarchical submodels of the full quadratic model. A hierarchical model contains an interaction only with both of its main effects, and a square only with its main effect. AICc, Akaike’s information criterion with the correction for small samples (Hurvich and Tsai 1989; Burnham and Anderson 2002), estimates how well a model would predict new runs. It rewards the fit and penalises every estimated parameter, and it counts the residual variance as one of them. Its correction raises the penalty when the runs are few for the number of parameters, as in a design of 28 runs, so that the criterion does not favour too many terms. Every candidate was fitted, 1,337 hierarchical models for 4 factors, and the one with the lowest AICc was taken.

A stepwise search, which adds or removes one term at a time, stops when no single change lowers AICc. It can therefore stop at a model that only a combination of changes would improve. On the data of this campaign a stepwise search by AICc would have stopped at a model with a higher AICc for 12 of the 21 outputs that have a criterion and an AICc. At this step a search forward from the mean would have stopped at 4 terms for pool host cell protein. A search backward from the full quadratic model would have stopped at 4 terms for leached Protein A and 5 for fragments, each with a higher AICc than the selected model. The backward elimination of INV-002 corrective action CA2 differed in three ways. It was stepwise, it removed terms by p-value, a test of each term against zero and not a comparison of models, and it used the raw scale only. For fragments it kept elution buffer pH alone, where the search keeps protein load and elution buffer pH.

The selected model was then judged against the model adequacy criteria of PCP-005 §7 listed below. A model with no term has no F-test, and criterion 2 does not apply to it. Such a model shows that, within the range studied, the output does not depend

on the parameters by more than the variation of the runs can resolve. Finally, the scale-down offset of each output was taken on the scale of its model.

3.5.3 Adequacy, prediction and ranges

Each model is reported with R^2, adjusted R^2, predicted R^2 (from PRESS) (Montgomery 2017), the model F-test, and the test of lack of fit against pure error from the centre points. Coefficients are reported in coded units, so that each measures the change in the response over half the range studied, in natural-log units for a log-scale model. The adequacy criteria of PCP-005 §7 were applied as follows. The numbers are those of PCP-005 §7, where criterion 1 is the qualification of the scale-down model (Section 3.1).

2. Model significance. The model F-test is significant at $\alpha = 0.05$.
3. Lack of fit. The lack of fit test is not significant at $\alpha = 0.05$.
4. Predictive power. Predicted R^2 falls short of R^2 by no more than 0.20. PCMP-001 §7.2 compares predicted R^2 with R^2, not with adjusted R^2.
5. Residuals. Regressions of the absolute residuals on the predicted values, and of the residuals on run order, show no slope significant at $\alpha = 0.05$. A run with a Cook's distance above 4/n (0.14 for 28 runs) is examined by deleting it and repeating the judgement. The model fails the criterion only where a run's Cook's distance exceeds the median of the F distribution with p and $n - p$ degrees of freedom, p being the number of coefficients of the model.

PCMP-001 §7.2 provides that a model that fails the lack of fit test, or whose predicted R^2 falls short of R^2 by more than 0.20, is refitted with a justified change of terms or transformation, or is not used to set a range.

Each output with a criterion is judged over the NOR at every combination of NOR low, set-point and NOR high of the 4 response-surface factors. The value compared with the criterion is the upper end of the 95 % prediction interval of a single run, which includes the uncertainty of the fitted coefficients and the residual. The criteria of this step are upper limits, so only that end is compared, which makes it a one-sided 97.5 % prediction limit (PCP-005 §7). This report calls it the upper prediction limit.

The PAR analysis follows PCMP-001 §6.4. The univariate PAR, which PCP-005 §8.2 calls the set-point analysis, holds the other parameters at their set-points. It is judged on the prediction of the model, the median for pool host cell protein and the mean for the other outputs. The NOR-propagated PAR is a multivariate PAR. It is the range of a parameter over which a single batch meets the criterion while the other parameters vary within their NORs. It is computed by Monte Carlo at 81 grid points across the range studied, with 2,000 operating points at each. Each other parameter is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of its NOR, clipped to its range studied, and the coefficients of the model are drawn from their estimated covariance together with the residual. At each grid point the 97.5th percentile of the simulated batches must meet an upper criterion, so that at least 97.5 % of them meet it. The NOR-propagated PAR is the one that supports a

PAR carried into the control strategy. The design space is drawn on the 95 % prediction interval of a single run, on a grid of 41 points per factor over the range studied (Section 6).

For each output with a criterion, each run with a Cook's distance above 4/n was deleted in turn, the model selection repeated and the output judged over the NOR again, as criterion 5 requires. Where a selected model does not meet criterion 5, the size of each residual trend is stated against the criterion of its output, and the use of that model for the ranges and the design space is the discrepancy PD-005-02 (Section 5.5).

Capability at commercial scale was estimated from the integrated process model of the train, which runs 2,000 simulated batches with each parameter drawn within its NOR. Capability is reported as Cpk against the drug substance criterion, one-sided for a criterion with one limit and the lower of the two one-sided indices for a two-sided criterion. It is judged against the target of 1.33 and the minimum of 1.00 of PCMP-001 §7.9. The same simulation counts the batches outside each drug substance criterion. Where none falls outside, the probability of an out-of-specification batch in the simulation is bounded by the one-sided 95 % upper confidence limit of a proportion with no event in 2,000 trials.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 lists the parameters of the step with their set-point, NOR, range studied, classification and study type.

Table 4.1: Parameters of the Protein A step with set-point, NOR, range studied, classification and study type.

Parameter	Unit	Set-point	NOR	Range studied	Class	Study
Protein load	g/L resin	30	20–40[a)]	10–50	WC-CPP	multivariate
Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	WC-CPP	multivariate
Load flow rate	cm/hr	200	150–250	100–300	KPP	multivariate
End of pool collect	CV	2.6	2.3–2.9	2–3.2	KPP	multivariate
Operating temperature	°C	22	18–26	15–30	GPP	univariate
Bed height	cm	20	15–25	10–30	GPP	univariate

[a)] Over this NOR the scale-down model is not qualified for pool HCP. The NOR is not narrowed, and it is flagged for discussion with manufacturing.

The execution window of every study of the step is recorded in Table 4.2. Each study began after PCP-005 Version 1.0 was approved on 2025-11-03. Study S-PA-UNI ran before the response-surface design began, against the order of PCP-005 §11, in which the univariate assessments come fourth (discrepancy PD-005-09, Section 13.3.5).

Table 4.2: Execution window of each study of the Protein A step.

Study	Study type	Start	End
S-PA-SDM	scale-down model qualification	2025-11-05	2025-11-14
S-PA-SCR	screening design	2025-11-17	2025-12-05
S-PA-UNI	univariate study	2025-12-08	2025-12-12
S-PA-RSM	response-surface design	2026-01-12	2026-02-06
S-PA-RET	retest of retained samples	2026-04-06	2026-04-17
S-PA-SDM2	scale-down model qualification	2026-09-07	2026-09-11
S-PA-HLD	hold study	2026-09-19	2026-09-28

4.2 Screening design

The screening design (study S-PA-SCR, 2025-11-17 to 2025-12-05) was a full two-level factorial design in the 4 factors, with 16 factorial runs and 3 centre points, 19 runs in all. In a full factorial no effect is aliased with another, and every main effect and every two-factor interaction can be estimated. The responses were pool host cell protein, step yield and leached Protein A, with fragments by SEC recorded for information. All 4 factors were carried into the response-surface design whatever the screening result, as PCP-005 §6.2 planned. Appendix A gives the matrix and the results.

Table 4.3 gives the letter code of each design factor, which the coefficient tables and the appendices use.

Table 4.3: Letter codes of the design factors.

Code	Factor	Unit
A	Protein load	g/L resin
B	Elution buffer pH	pH
C	Load flow rate	cm/hr
D	End of pool collect	CV

4.3 Response-surface design

The response-surface design (study S-PA-RSM, 2026-01-12 to 2026-02-06) was a face-centred central composite design in the same 4 factors. It has 16 factorial runs, 8 axial runs at the faces of the cube and 4 centre points, 28 runs in all. The full quadratic model, an intercept and 14 terms, would leave 13 residual degrees of freedom, with 3 for pure error and 10 for lack of fit. The responses were pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS on the retained pools. Appendix B gives the matrix and the results.

4.4 Univariate assessment

Operating temperature and bed height were studied one factor at a time in the univariate study S-PA-UNI (2025-12-08 to 2025-12-12), with the other parameters at their set-points. Each was run 2 times at the low and the high end of its range studied. The 3 set-point runs of the planned qualification study S-PA-SDM served as the reference, in place of the runs at the centre of each range that PCP-005 §6.4 requires (discrepancy PD-005-07, Section 13.3.3). Bed height was varied at the set-point linear velocity, so that the residence time changed with it. The shortest bed gives a residence time of 3 min, shorter than any run of the designs. As PCP-005 §6.4 requires, the responses were those of the multivariate designs, pool host cell protein, step yield and leached Protein A, with aggregate and fragments by SEC. Fragments were not measured by nrCE-SDS (Section 3.3). For pool host cell protein and leached Protein

A, the range of operating temperature or of bed height is supported when every individual result at both ends meets its criterion at manufacturing scale (PCP-005 §6.4). A univariate study cannot detect an interaction. The ranges of these two parameters are therefore supported only with the other parameters at their set-points.

4.5 Hold study of the Protein A eluate

The Protein A eluate is held at elution pH before viral inactivation. The hold study S-PA-HLD (2026-09-19 to 2026-09-28) was made during the Version 2.0 re-analysis, after INV-001 closed, and is recorded as the discrepancy PD-005-05 (Section 13.3.2). One eluate was produced at set-point, with elution buffer pH 3.55, divided and held at 2-8 °C and 21 °C for up to 7 days. Samples were taken on days 0, 1, 3, 5 and 7 at each temperature, and aggregate by SEC, acidic variants by icIEF and fragments by nrCE-SDS were measured at every sample point.

The plan set no hold criterion, because it foresaw no hold study. The held eluate is the load of low-pH viral inactivation, so it is judged against the intermediate acceptance criteria in force for that load, as the verification of PCMP-001 §7.5 left them, and not against their a priori values. For aggregate and acidic variants they equal the criteria of the production bioreactor output, because capture is expected to leave both unchanged (Section 2), and for fragments that of the Protein A pool. This report does not test the expectation for aggregate and acidic variants, because neither was measured in the load of this step. For aggregate the criterion is ≤ 6.30 % HMW. It is the drug substance limit of 5 % less the aggregate increase of UF/DF, 0.104 percentage points, multiplied by the aggregate clearance of cation exchange, 1.398-fold, less the aggregate increase of low-pH viral inactivation, 0.538 percentage points. Each value is the one-sided 95 % prediction bound of a single batch of its step, which replaced a platform worst case (PCMP-001 §7.5). PCMP-001 assumed that no step after cation exchange changes aggregate. UF/DF raises it within its NOR, which made the criterion stricter. Cation exchange cleared less aggregate within its NOR than its platform worst case of 2.00-fold, which made the criterion stricter. Low-pH viral inactivation added less aggregate than its platform worst case of 0.70 points, which made it wider. The net change, from ≤ 9.30 % a priori to ≤ 6.30 %, makes the criterion stricter. The platform aggregate increase was measured over a hold of 180 min, the longest hold studied at low pH, and the A-Mab bound within the hold-time NOR as characterized, which ends at 120 min. The bound is smaller because the A-Mab hold is shorter, not because the product is more stable. At the edges of the range studied A-Mab added up to 0.82 points of aggregate, more than the platform value of 0.70, and within the NOR at most 0.50 points. The aggregate criterion is still right, because routine manufacture does not hold longer than the NOR.

For acidic variants the criterion is 18–39 %, its a priori value. Its upper limit is the drug substance limit of 40 % less the platform worst case of the increase in low-pH viral inactivation, 1.00 percentage points (Section 7), and its lower limit is that of the drug substance. Fragments are judged against the intermediate acceptance criterion of the pool, ≤ 2.45 % by nrCE-SDS.

Against these criteria the report judges what the hold adds to the eluate, by a hold rule of its own that is part of the discrepancy PD-005-05. A hold time is supported at a temperature when the largest increase from day 0 up to that time, added to the highest level expected in the Protein A pool in manufacture, stays within its criterion. For aggregate and acidic variants the highest level expected is the upper prediction limit of the production bioreactor model over its NOR (PCR-003), and for fragments that of the fragment model of this step, 1.09 % (Section 5.3.4). Acidic variants are also judged against their lower limit (INV-001 finding 4).

5 Results

5.1 Centre-point performance and reproducibility

The centre points of each design were run at set-point and estimate the pure error against which lack of fit is tested. Table 5.1 gives their mean, standard deviation and coefficient of variation in the response-surface design, and Table 5.2 in the screening design. Fragments by nrCE-SDS were measured in the response-surface design only.

Table 5.1: Centre-point reproducibility in the response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	7,564	357	4.7
Step yield (%)	4	94.6	1.87	2.0
Leached Protein A (ppm)	4	2.86	0.97	33.8
Fragments, nrCE-SDS (%)	4	0.89	0.08	8.7

Table 5.2: Centre-point reproducibility in the screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	8,103	1,611	19.9
Step yield (%)	3	93.5	0.60	0.6
Leached Protein A (ppm)	3	3.28	0.51	15.5

Step yield was the most reproducible response, with a coefficient of variation of 2.0 % in the response-surface design. Pool host cell protein reproduced to 4.7 % in the response-surface design and to 19.9 % in the screening design. The larger screening value makes the screening tests for pool host cell protein less sensitive than those of the response-surface design. Fragments by nrCE-SDS reproduced to 8.7 %. Leached Protein A was the least precise response, at 33.8 % in the response-surface design. Its standard deviation at the centre points, 0.97 ppm, is 44 % larger than its standard deviation across all 28 runs of the design, 0.67 ppm. With pure error this large against the variation between runs, the lack of fit test of leached Protein A has little power.

5.2 Screening: factor effects

Table 5.3 summarizes the screening models. The models for pool host cell protein and step yield are significant. The model for leached Protein A is not (p = 0.867), and none of its terms is significant.

Table 5.3: Screening model summary (main effects and two-factor interactions).

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.973	0.939	0.886	28.65	< 0.001	1,400
Step yield (%)	19	0.976	0.945	0.798	32.04	< 0.001	0.83
Leached Protein A (ppm)	19	0.371	−0.414	−2.981	0.47	0.867	0.51

RMSE, root mean square error, for step yield in percentage points.

Pool host cell protein was affected by elution buffer pH (B) and protein load (A). Raising the elution pH across its range lowered pool host cell protein by 8,840 ng/mg, and raising protein load across its range raised it by 7,269 ng/mg. In ng/mg the screening model also shows an interaction of protein load with elution buffer pH (AB, −2,587 ng/mg, p = 0.006), so the fall with rising pH is larger at high load. Two effects that multiply produce an interaction of this sign in ng/mg, and the response-surface analysis examines it on the log scale (Section 5.3.1). Load flow rate and end of pool collect had no significant effect in the screening design (Table 5.4).

Table 5.4: Screening effects on pool HCP (ng/mg), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value
B	−8,840	−4,420	350	−12.63	< 0.001
A	7,269	3,635	350	10.38	< 0.001
AB	−2,587	−1,293	350	−3.70	0.006
AD	1,133	566	350	1.62	0.144
AC	830	415	350	1.19	0.270
D	708	354	350	1.01	0.341
BD	−433	−217	350	−0.62	0.553
C	167	84	350	0.24	0.817
CD	−71	−35	350	−0.10	0.922
BC	6	3	350	0.01	0.993

Step yield was affected by end of pool collect (D), protein load (A) and load flow rate (C), and by the interaction of protein load with load flow rate (AC). Extending the end of pool collect across its range raised yield by 6.0 percentage points. Raising protein load and load flow rate lowered it by 2.6 and 2.4 percentage points. The interaction of protein load with load flow rate (−2.1 percentage points, p = 0.001) means that flow rate costs more yield at high load. The interactions of load flow rate with end of pool collect (CD) and of elution buffer pH with end of pool collect (BD) were also significant, at −1.1 and −1.0 percentage points. They are smaller than the significant main effects, and neither is in the model selected for step yield in the

response-surface design. The p-value of BD, 0.046, rests on the unrounded results. Refitted from the yields as Appendix A prints them, to 0.1 %, it is 0.052. Elution buffer pH had no significant effect on yield. Table 5.5 gives the effects, and Appendix C gives the effects for leached Protein A.

Table 5.5: Screening effects on step yield (percentage points), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value
D	6.04	3.02	0.21	14.50	< 0.001
A	−2.55	−1.28	0.21	−6.13	< 0.001
C	−2.37	−1.18	0.21	−5.68	< 0.001
AC	−2.09	−1.04	0.21	−5.01	0.001
CD	−1.07	−0.54	0.21	−2.58	0.033
BD	−0.98	−0.49	0.21	−2.36	0.046
AB	0.44	0.22	0.21	1.06	0.319
AD	0.37	0.19	0.21	0.89	0.400
BC	0.35	0.17	0.21	0.84	0.425
B	−0.24	−0.12	0.21	−0.57	0.586

Computed from the unrounded results. Appendix A prints step yield to 0.1 %.

The screening results agree in direction with the expectations of Section 2 for protein load and elution buffer pH on pool host cell protein, and for protein load, load flow rate, their interaction and end of pool collect on step yield.

5.3 Response-surface models

Each output of the response-surface design was analysed by the model selection of Section 3.5. Under the rule of the investigation INV-002 for planned models that failed PCMP-001 §7.2, no runs were added to the Protein A design. Leached Protein A showed no effect to resolve, and for fragments no second block within the size of the planned design reached the assurance of at least 0.90 that INV-002 requires. Every run of the design is therefore in one block. Table 5.6 compares the full quadratic model that PCP-005 planned, fitted to the 28 planned runs, with the model selected for each output with a criterion.

In the full quadratic model the predicted R^2 of leached Protein A and of fragments is below zero. Both fail by a wide margin the limit of PCMP-001 §7.2 that predicted R^2 may fall no more than 0.20 below R^2. For pool host cell protein the full quadratic model on the raw scale, with run 23 at its recorded pH, fails the lack of fit test ($p = 0.024$). With run 23 at its design pH, as Version 1.0 and INV-002 fitted it, the same model meets both tests, with a gap of 0.057 and a lack of fit test of $p = 0.058$. The full quadratic model of step yield, which has no criterion and is not in the table, also fails the limit, with predicted R^2 0.31 below R^2.

Table 5.6: Re-evaluation of the response-surface models.

Attribute	Planned runs, full quadratic				Model used (terms)
	Runs	R^2	Pred. R^2	Gap	
Host cell protein (ng/mg)	28	0.98	0.89	0.09	5 terms, log scale
Leached Protein A (ppm)	28	0.57	−0.74	1.30	No effect
Fragments, LMW (%)	28	0.61	−1.28	1.89	Protein load, elution buffer pH

Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, on one block of runs (INV-002).
At this step the full quadratic model is on the raw scale, with run 23 at its recorded elution buffer pH (PD-005-03). The design was not augmented and has no block term. No effect, a selected model with the intercept only.

Table 5.7 judges each selected model against criteria 2, 3 and 4 of PCP-005 §7.

Table 5.7: Selected response-surface models against criteria 2, 3 and 4 of PCP-005 §7.

Output	Scale	Terms	R^2	Adj. R^2	Pred. R^2	Criterion 2		Criterion 3		Criterion 4	
						Model p	Met	LoF p	Met	Gap	Met
Pool HCP (ng/mg)	log	A, B, C, D, C^2	0.977	0.971	0.963	< 0.001	Yes	0.088	Yes	0.014	Yes
Step yield (%)	raw	A, C, D, AC, D^2	0.862	0.831	0.782	< 0.001	Yes	0.771	Yes	0.081	Yes
Leached Protein A (ppm)	raw	none	0.000	0.000	−0.075	n.a.	n.a.	0.907	Yes	0.075	Yes
Fragments, nrCE-SDS (%)	raw	A, B	0.396	0.348	0.235	0.002	Yes	0.776	Yes	0.161	Yes

Terms, the terms of the selected model besides the intercept. A, protein load. B, elution buffer pH. C, load flow rate. D, end of pool collect. C^2, the square of C. AC, the interaction of A with C. Criterion 2, the model F-test significant at $\alpha = 0.05$ (n.a., a model with no term, which has no F-test). Criterion 3, the lack of fit (LoF) test against pure error not significant at $\alpha = 0.05$. Criterion 4, the gap, R^2 minus predicted R^2, at most 0.20.

Each selected model meets criteria 3 and 4, and each model with a term meets criterion 2. The selection kept 5 of the 14 terms of the full quadratic model for pool host cell protein, 5 for step yield, 2 for fragments and none for leached Protein A.

5.3.1 Pool host cell protein

Table 5.8 gives the coefficients of the selected model for pool host cell protein, on the natural log scale, and Table 5.9 its analysis of variance.

Table 5.8: Response-surface coefficients for pool HCP, natural log of ng/mg, coded units.

Term	Coef.	Std. err.	t	p-value
Intercept	8.848	0.033	265.73	< 0.001
A	0.443	0.025	17.87	< 0.001
B	−0.596	0.025	−24.17	< 0.001
C	−0.027	0.025	−1.09	0.288
D	−0.082	0.025	−3.30	0.003
C^2	0.100	0.042	2.40	0.025

Table 5.9: Analysis of variance with lack of fit for pool HCP, natural log of ng/mg.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	10.2228	5	2.0446	184.65	< 0.001
Residual	0.2436	22	0.0111		
Lack of fit	0.2370	19	0.0125	5.70	0.088
Pure error	0.0066	3	0.0022		

Protein load and elution buffer pH carry the largest coefficients, 0.443 and −0.596 per half range. Across the range studied pool host cell protein rises by a factor of 2.43 with protein load, and it falls by a factor of 3.30 as elution buffer pH rises from 3.2 to 3.9. On the log scale the two effects add, so in ng/mg they multiply. With load flow rate and end of pool collect at their set-points, raising the pH across its range lowers pool host cell protein by 14,081 ng/mg at a protein load of 50 g/L resin and by 5,804 ng/mg at 10 g/L resin, at manufacturing scale. The factor is the same at both loads. In ng/mg the fall is larger at high load because the level it acts on is higher.

A model on the raw scale can describe two effects that multiply only through an interaction. In the full quadratic model on the raw scale the interaction of protein load with elution buffer pH is significant ($p < 0.001$), and the model of Version 1.0 carried it. In the full quadratic model on the log scale it is not significant ($p = 0.156$), and the selected model does not contain it. The raw-scale interaction was therefore largely the multiplicative structure of the two main effects. Whether a further interaction of the two exists on the log scale the data do not decide. Its p-value is above 0.05 in each of the three analyses of run 23, and the selection keeps it in 2 of the 3 (Section 13.1). Table 5.10 lists the 9 models whose AICc lies within 2 units of that of the selected model. All of them contain protein load, elution buffer pH and end of pool collect. They differ in load flow rate, in the interaction of protein load with elution buffer pH, and in the squares of the four factors. The interaction is in 4 of them, the nearest 1.06 units from the selected model, and its p-value is at least 0.126 in each. Models within 2 units of the lowest AICc have substantial support from the data (Burnham and Anderson 2002), so the data support each of these models about as well as the selected one.

Table 5.10: Models of pool HCP within 2 AICc units of the selected model, log scale, with the upper prediction limit over the NOR.

Terms	ΔAICc	Upper prediction limit over the NOR, ng/mg		Criteria 2 to 4 met	Scale-down model qualified
		At the estimate	At the low end		
A, B, C, D, C^2	0.00	15,697	17,143	Yes	No
A, B, C, D, A^2, C^2	0.33	15,743	17,193	Yes	No
A, B, D, B^2	0.38	15,667	17,109	Yes	No
A, B, D	0.82	16,374	17,882	Yes	No
A, B, C, D, AB, C^2	1.06	15,480	16,905	Yes	No
A, B, C, D, AB, A^2, C^2	1.40	15,507	16,935	Yes	No
A, B, D, AB, B^2	1.46	15,467	16,891	Yes	No
A, B, D, AB	1.91	16,182	17,672	Yes	No
A, B, C, D, C^2, D^2	1.94	15,814	17,271	Yes	No

Terms of the model besides the intercept, coded as in the factor legend. AB, the interaction of A with B. A^2, B^2, C^2, D^2, the squares of A, B, C and D. ΔAICc, the AICc of the model minus that of the selected model, the first row. AICc counts the residual variance as a parameter. At the estimate and at the low end, at manufacturing scale over the NOR in force, with the offset ratio at its estimate and at the low end of its 90 % confidence interval. Scale-down model qualified, both limits within the criterion of ≤ 16,719 ng/mg.

Each of these models meets criteria 2 to 4. With the offset ratio at its estimate, each gives an upper prediction limit over the NOR between 15,467 and 16,374 ng/mg at manufacturing scale, below the criterion of ≤ 16,719 ng/mg, so the conclusion over the NOR is the same whichever model is used. With the ratio at the low end of its interval each of the 9 models exceeds the criterion, with limits between 16,891 and 17,882 ng/mg. With none of them is the scale-down model qualified for pool host cell protein over the NOR in force (Section 3.1). That conclusion therefore does not depend on the choice of model.

End of pool collect lowers pool host cell protein. Its coefficient of −0.082 per half range ($p = 0.003$) is a fall of 15 % across the range studied. Load flow rate has no significant linear effect ($p = 0.288$). Its square is significant ($p = 0.025$), and pool host cell protein is higher at both ends of the flow range than at the set-point, by 8 % at 300 cm/hr and 14 % at 100 cm/hr. At the edges of the NOR of load flow rate the rise is at most 4 %. The lack of fit test ($p = 0.088$) is not significant.

The effect of end of pool collect has opposite signs in the two designs. In the screening design it was 708 ng/mg across its range ($p = 0.341$), and in the response-surface design it is a fall of 15 %. The screening estimate is not significant, with a standard error of 700 ng/mg. Noise alone does not explain the difference, however. The screening results were compared with the predictions of the response-surface model at the same settings. Their ratio rises with end of pool collect, by a factor of 1.22 across its range ($p = 0.021$). Curvature does not explain it either, because in a full factorial design the quadratic terms are confounded with the intercept, not with the main effect of end of pool collect. The cause was not identified (finding F11). A feed that carries more host cell protein in the tail of the elution peak would give the screening sign, because a later end of collection would then add host cell protein to the pool. The

two designs were run at different times (Table 4.2), and their feeds were not characterized for this. The effect of end of pool collect is taken from the response-surface model, which is the predictive model.

Across the response-surface design pool host cell protein ranged from 2,149 to 25,114 ng/mg in the scale-down model. At manufacturing scale, 8 results of the two designs exceeded the criterion of ≤ 16,719 ng/mg, at 17,021 to 25,800 ng/mg. Each of them was at the highest protein load and the lowest elution buffer pH of the range studied, outside both NORs.

Figure 5.1 shows the response surfaces of the scale-down model over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points. In the pool host cell protein panel the contours are straight and parallel, because the selected model has no interaction of protein load with elution buffer pH. The contour levels are evenly spaced in ng/mg. Because the effects multiply, the bands between them are narrower where pool host cell protein is high, at high load and low pH. The leached Protein A panel is flat, because its selected model has no term.

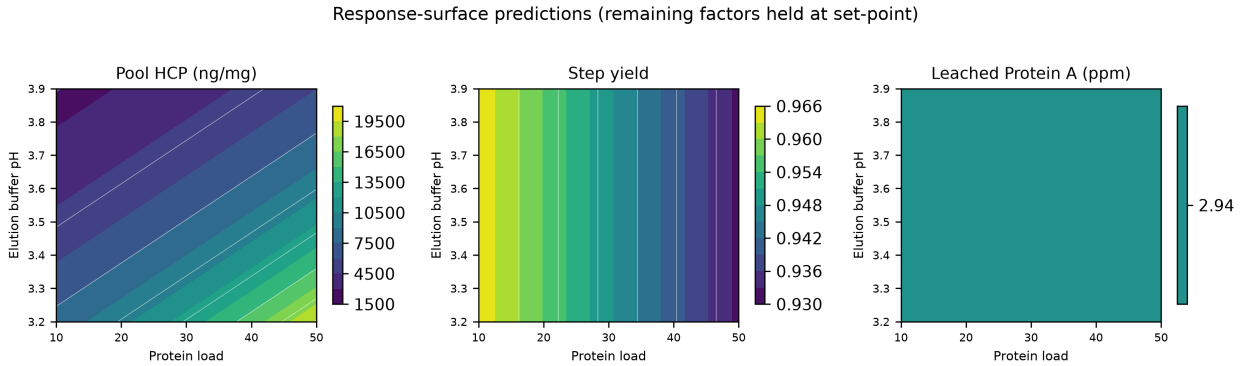


Figure 5.1: Response surfaces of the scale-down model over protein load and elution buffer pH, with load flow rate and end of pool collect at their set-points (the design centre). Pool HCP, the median of the log-scale model. Step yield as a fraction. Leached Protein A, the mean of the runs, since its model has no term.

Figure 5.2 shows the diagnostics of the model on the log scale. The residuals scatter about zero across the predicted range without a funnel. On the normal probability plot they follow the line in the middle, and the largest positive residuals fall below it, so the upper tail of the residuals is shorter than that of a normal distribution. The lowest residual belongs to run 23, at 2.0 residual standard deviations below its prediction (Section 13.1).

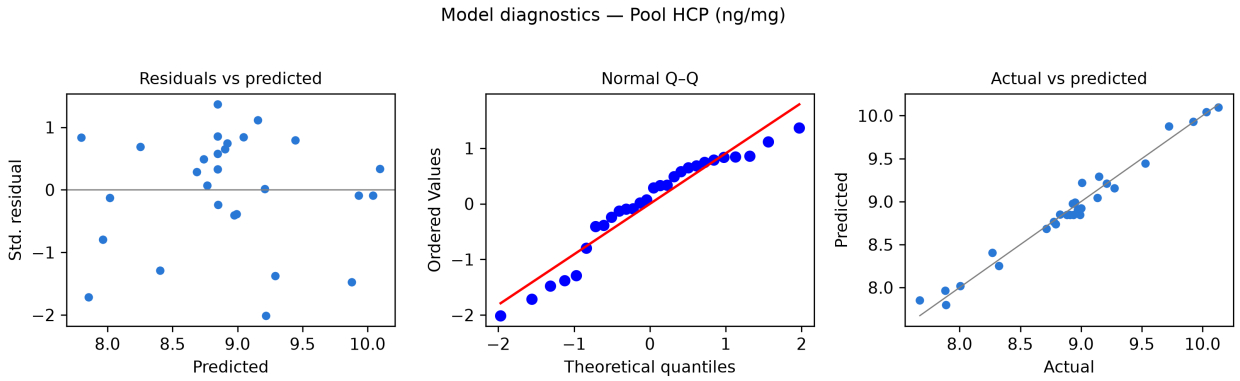


Figure 5.2: Diagnostics of the selected response-surface model for pool HCP, on the natural log scale.

5.3.2 Step yield

Table 5.11 gives the coefficients of the selected model for step yield, in percentage points.

Table 5.11: Response-surface coefficients for step yield (percentage points), coded units.

Term	Coef.	Std. err.	t	p-value
Intercept	94.86	0.49	194.20	< 0.001
A	−1.65	0.36	−4.52	< 0.001
C	−0.89	0.36	−2.45	0.023
D	3.65	0.36	10.04	< 0.001
AC	−0.91	0.39	−2.37	0.027
D^2	−1.37	0.61	−2.25	0.035

End of pool collect has the largest effect. Extending it from 2 to 2.6 CV raises yield by 5.0 percentage points, and from 2.6 to 3.2 CV by a further 2.3 points. The negative square of end of pool collect makes the gain smaller at the upper end of its range. Protein load lowers yield by 1.65 points per half range. At high load the coefficient of load flow rate is −1.80 points per half range, and at low load it is 0.02. Flow rate therefore has almost no effect on yield when the bed has capacity to spare. Across the response-surface design step yield ranged from 84.4 to 99.0 %. Figure 5.3 shows the diagnostics of the model.

In the designs and the univariate study, 6 results of step yield are exactly 99.0 %, and none is higher. Of these, 5 were at the upper end of the range of end of pool collect, 3.2 CV. These results may reflect a limit on the reported yield. If they do, the gain at the upper end of end of pool collect is understated, and part of the negative square of end of pool collect may come from that limit.

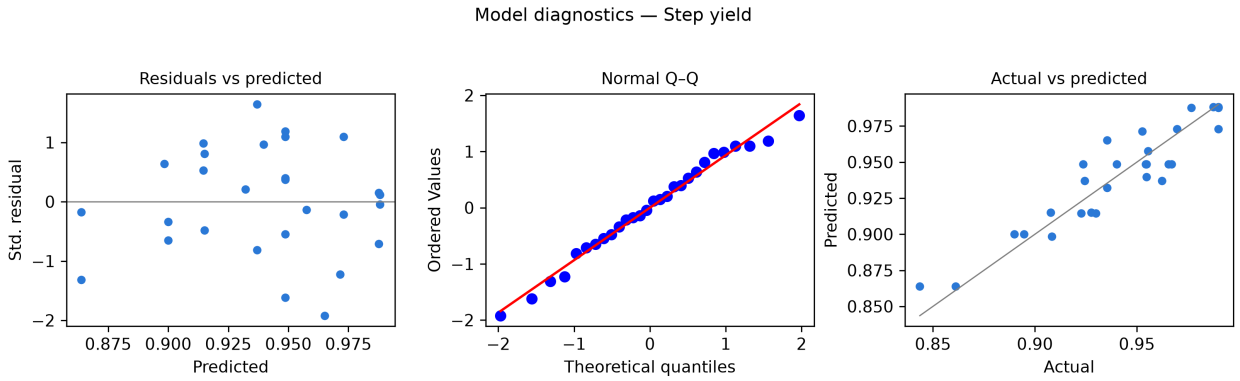


Figure 5.3: Diagnostics of the selected response-surface model for step yield (as a fraction).

5.3.3 Leached Protein A

The selected model of leached Protein A has no term. Within the range studied, leached Protein A does not depend on the 4 parameters of the multivariate design by more than the design resolves. With the 28 runs and the scatter of this output, the half-width of the 95 % confidence interval of a main effect across the range studied is 0.67 ppm in a model of the four main effects. An effect smaller than this would not be told apart from zero. The model is the mean of the runs, 2.94 ppm in the scale-down model and 4.12 ppm at manufacturing scale. In the scale-down model leached Protein A ranged from 1.70 to 4.61 ppm across the runs of both designs. At manufacturing scale the highest result corresponds to 5.79 ppm, and every run met the intermediate acceptance criterion of ≤ 158 ppm (finding F5). These results use the re-assay of run 11. With the first result of that run in its place, the selection keeps end of pool collect, and that model meets criteria 2 to 4 (Section 13.2).

5.3.4 Fragments by nrCE-SDS

Table 5.12 gives the coefficients of the selected model for fragments by nrCE-SDS, measured on the 28 retained pools of the response-surface design.

Table 5.12: Response-surface coefficients for fragments by nrCE-SDS (%), coded units.

Term	Coef.	Std. err.	t	p-value
Intercept	0.921	0.012	76.44	< 0.001
A	−0.028	0.015	−1.83	0.079
B	−0.054	0.015	−3.61	0.001

The model is significant (p = 0.002), shows no significant lack of fit (p = 0.776), and its predicted R^2 falls short of R^2 by 0.161, within the limit of 0.20. Its R^2 of 0.396

shows that most of the variation between pools is not explained by the parameters. Fragments fall as elution buffer pH rises, by 0.11 percentage points across the range studied (p = 0.001). The selection also kept protein load, whose coefficient is not significant at α = 0.05 (p = 0.079). AICc keeps a term when the improvement of the fit outweighs the penalty for its coefficient, and such a term need not be significant at α = 0.05. The effect of protein load on fragments is not interpreted. Over the NOR the upper prediction limit of a single run is 1.09 %, against the intermediate acceptance criterion of ≤ 2.45 %. Fragments ranged from 0.77 to 1.09 % across the pools, and every pool met the criterion. These results are in the scale-down model, because fragments carry no offset (Section 3.1).

5.4 Attributes measured for information

Aggregate by SEC was measured on every pool of both designs, for information. It ranged from 1.40 to 1.73 % HMW across the 47 pools, below the ≤ 6.30 % that applies to the held eluate. Aggregate was not measured in the load, so these results do not show whether capture changes it. Appendix A and Appendix B give the individual results.

Residual DNA was measured on the load and the pool of the 7 centre-point runs of the two designs. With the antibody in the load measured by Protein A affinity HPLC, the load carried 137,045 to 234,778 pg per mg antibody, and the pool 498 to 1,372 pg/mg. That is a reduction of 2.2 to 2.5 log10 across the step. DNA was not measured away from the set-point, as PCP-005 planned, so its dependence on the parameters is not tested here.

5.5 Model residuals and influential runs

Table 5.13 gives the residual checks of criterion 5 of PCP-005 §7 for the selected models.

Table 5.13: Residual checks of the selected response-surface models (PCP-005 §7, criterion 5).

Output	Trend on predicted, p	Trend on run order, p	Largest Cook's D	Run with largest D	Runs above 4/n	F median	Met
Pool HCP (ng/mg)	0.463	0.586	0.26	6	3	0.92	Yes
Step yield (%)	0.758	0.634	0.25	16	2	0.92	Yes
Leached Protein A (ppm)	n.a.	0.046	0.24	7	1	0.47	No
Fragments, nrCE-SDS (%)	0.021	0.872	0.17	1	1	0.81	No

Trend on predicted, the p-value of the regression of the absolute residuals on the predicted values (n.a., a model with no term predicts a single value). Trend on run order, the p-value of the regression of the residuals on the run order of execution, which Appendix B gives. Cook's D, Cook's distance. Runs above 4/n, the number of runs with a Cook's distance above 0.14, each examined by deleting it. F median, the median of the F distribution with p and n − p degrees of freedom, p being the number of coefficients of the model, above which a run fails it. Met, no trend significant at α = 0.05 and no run above the F median.

Two of the 7 trend tests are significant at α = 0.05, so criterion 5 is not met for leached Protein A or for fragments on their trends. The residuals of leached Protein A rise with the run order of execution (p = 0.046), by 0.84 ppm over the 28 runs. That is 0.5 % of the criterion of ≤ 158 ppm. The cause of the drift was not investigated. The absolute residuals of fragments rise with the predicted value (p = 0.021), so the spread of the fragment results grows with their level. Across the range of the predictions the mean absolute residual grows by 0.04 percentage points. The upper prediction limit of fragments over the NOR lies 1.36 points below its criterion.

Every selected model has runs above the Cook's distance limit of 4/n, between 1 and 3 per output. The cut-off flags runs even when a model is correct. In data simulated from each selected model with normal residuals, it flags on average between 1.3 and 2.5 of the 28 runs, and at least one run in 87 to 96 % of the data sets. A flagged run is therefore examined, and it fails the model only where its Cook's distance exceeds the median of F(p, n − p), 0.47 to 0.92 for these models. No run exceeds it, and the largest Cook's distance is 0.26. Criterion 5 is therefore met for pool host cell protein and step yield, and it is not met for leached Protein A and fragments, because of their residual trends.

For each output with a criterion, each run above 4/n was deleted in turn, the model selection repeated, and the output judged over the NOR again. For pool host cell protein these are runs 6, 12 and 23. With any one of them deleted, the upper prediction limit over the NOR lies between 15,308 and 15,554 ng/mg at manufacturing scale, against 15,697 ng/mg with every run. Each of these is below the criterion of ≤ 16,719 ng/mg. With the offset at the low end of its interval the limit is 16,717 ng/mg with run 6 deleted, just within the criterion, and up to 16,987 ng/mg with either of the other runs deleted, as it is above the criterion with every run (Section 3.1). Deleting a run is a test of its influence, and the qualification is judged on all runs. The coefficients of protein load and elution buffer pH change by at most 3.1 %. Deleting run 23 adds the interaction of protein load with elution buffer pH and the square of

protein load to the selected model (Section 13.1). Run 23 therefore decides whether the selected model contains that interaction, but not the conclusion over the NOR. Leached Protein A with run 7 deleted, and fragments with run 1 deleted, still meet their criteria at every combination of the NOR.

5.6 Univariate assessment

Table 5.14 gives the results of the univariate study S-PA-UNI for pool host cell protein, step yield and leached Protein A. Each end of each range is the arithmetic mean of 2 runs. The difference is taken from the arithmetic mean of the 3 set-point runs of the planned qualification study S-PA-SDM, with its 95 % confidence interval (Student t, pooled variance). The qualification runs are the reference because the univariate study made no run at the centre of each range (PD-005-07, Section 13.3.3). Both are in the scale-down model. The offset of leached Protein A is a difference, and it cancels from the difference. The offset of pool host cell protein is a ratio, and at manufacturing scale each difference in pool host cell protein would be divided by it, a change of 3 %. Table 13.3 gives the geometric mean of pool host cell protein for the same set-point runs, because that output is modelled on the log scale.

Table 5.14: Univariate study of operating temperature and bed height (S-PA-UNI), difference from the set-point runs.

Parameter	Setting	HCP (ng/mg)		Yield (points)		LPA (ppm)	
		Δ	95 % CI	Δ	95 % CI	Δ	95 % CI
Operating temperature	15 °C	2,005	−281 to 4,291	3.6	−4.9 to 12.0	0.26	−1.46 to 1.99
	30 °C	417	−1,186 to 2,021	4.1	−4.4 to 12.5	1.51	−0.67 to 3.70
Bed height	10 cm	13	−1,446 to 1,471	3.1	−6.9 to 13.0	0.76	−1.12 to 2.65
	30 cm	486	−1,843 to 2,815	2.4	−7.8 to 12.6	0.59	−1.15 to 2.32

Δ, the arithmetic mean of the 2 runs at the setting minus the arithmetic mean of the 3 set-point runs of study S-PA-SDM, both in the scale-down model. CI, confidence interval. HCP, pool host cell protein. LPA, leached Protein A. Yield in percentage points. Aggregate and fragments by SEC have no difference, because the set-point runs of study S-PA-SDM were not tested by SEC. Fragments by nrCE-SDS were not tested in this study.

Table 5.15 gives the individual results, as measured in the scale-down model, with the reference runs.

Table 5.15: Individual results of study S-PA-UNI and of its reference runs, scale-down model.

Parameter	Setting	Run	HCP (ng/mg)	Yield (%)	LPA (ppm)	HMW, SEC (%)	LMW, SEC (%)
Operating temperature	15 °C	1	10,209	96.5	2.51	1.59	1.42
		2	8,628	97.8	2.22	1.55	1.34
	30 °C	1	7,441	98.4	3.03	1.56	1.24
		2	8,222	96.9	4.20	1.53	1.31
Bed height	10 cm	1	7,698	94.4	2.52	1.60	1.26
		2	7,155	99.0	3.21	1.50	1.26
	30 cm	1	8,712	93.5	2.84	1.57	1.40
		2	7,088	98.5	2.54	1.49	1.28
Reference, S-PA-SDM	set-point	1	7,436	89.9	2.88	n.t.	n.t.
		2	7,955	96.8	1.48	n.t.	n.t.
		3	6,852	94.1	1.94	n.t.	n.t.

HCP, pool host cell protein. LPA, leached Protein A. HMW, aggregate by SEC. LMW, fragments by SEC, biased in this matrix and reported for information only under the method decision AMD-001. Reference, the set-point runs of the planned qualification study S-PA-SDM. n.t., not tested. Fragments by nrCE-SDS were not tested in this study.

At manufacturing scale every individual result of pool host cell protein and leached Protein A met its criterion. The highest pool host cell protein was 10,488 ng/mg against ≤ 16,719 ng/mg, and the highest leached Protein A was 5.38 ppm against ≤ 158 ppm. For these two attributes the range studied of each parameter is supported with the other parameters at their set-points. The study gives no fragment result by nrCE-SDS, and it supports no range of operating temperature or bed height for fragments (Section 9).

Every confidence interval includes zero. The largest difference in pool host cell protein, 2,005 ng/mg at 15 °C of operating temperature, is not significant. With 2 runs at each end the intervals are wide, so the study excludes only effects larger than their upper bounds. Step yield was 2.4 to 4.1 percentage points higher than the reference at every end. The difference has the same sign at both ends of both parameters. It is therefore more likely a difference between the reference runs and the univariate runs than an effect of either parameter (Section 13.3.3).

Aggregate by SEC ranged from 1.49 to 1.60 % HMW across the 8 runs of the study, within the range of the pools of both designs and below the ≤ 6.30 % that applies to the held eluate. Fragments by SEC ranged from 1.24 to 1.42 %. In the Protein A pool SEC reads part of the host cell protein as fragment, so this result is biased and is reported for information only (AMD-001). Neither SEC result has a difference from the set-point, because the reference runs of study S-PA-SDM were not tested by SEC.

5.7 Hold study of the Protein A eluate

Table 5.16 gives the results of the hold study.

Table 5.16: Hold-time stability of the Protein A eluate (held at elution pH).

Storage	Day	Size variants (%)		Acidic variants (%)
		HMW (SEC)	LMW (nrCE-SDS)	
2-8 °C	0	1.61	0.91	21.7
	1	1.57	0.86	22.0
	3	1.63	0.88	21.9
	5	1.66	0.93	21.8
	7	1.72	0.99	22.0
21 °C	0	1.54	0.83	22.3
	1	1.71	0.79	22.2
	3	1.80	1.05	22.4
	5	1.77	0.99	22.4
	7	1.95	1.08	22.6

The study held one eluate, made at set-point with elution buffer pH 3.55, and the two temperatures started from aliquots of it. On day 0 these aliquots already differed by 0.07 percentage points in aggregate, 0.6 points in acidic variants and 0.08 points in fragments by nrCE-SDS. This is the variation of the methods and of the aliquots, and a change smaller than it is not resolved.

Over 7 days aggregate rose by 0.41 percentage points at 21 °C and 0.11 points at 2-8 °C. The faster rise when warm is consistent with partial unfolding at low pH. Fragments by nrCE-SDS rose by 0.25 points at 21 °C, more than the 0.08 points between the day 0 aliquots, which is consistent with faster hinge hydrolysis when warm. At 2-8 °C they changed by 0.084 points, about the same as the 0.081 points between the day 0 aliquots, which is not resolved. Acidic variants changed by 0.3 points at both temperatures, less than the 0.6 points between the day 0 aliquots. The hold therefore did not change acidic variants by more than the methods and aliquots resolve.

Table 5.17 judges the hold against the criteria of the low-pH inactivation load by the rule of Section 4.5, and acidic variants also against their lower limit.

Table 5.17: Hold of the Protein A eluate up to day 7 against the criteria of the low-pH inactivation load, with acidic variants also against their lower limit.

Attribute	Limit	Criterion	Level entering the hold	Temperature	Change over the hold	Level after the hold	Met
Aggregate, HMW (%)	Upper	≤ 6.30	2.35	2-8 °C	+0.11	2.46	Yes
				21 °C	+0.41	2.76	Yes
Acidic variants (%)	Upper	≤ 39.0	30.3	2-8 °C	+0.3	30.6	Yes
				21 °C	+0.3	30.6	Yes
	Lower	≥ 18	16.2	2-8 °C	0.0	16.2	No[a)]
				21 °C	−0.1	16.1	No[a)]
Fragments, nrCE-SDS (%)	Upper	≤ 2.45	1.09	2-8 °C	+0.08	1.17	Yes
				21 °C	+0.25	1.34	Yes

[a)] Below the lower limit before the hold, at the least favourable corner of the production bioreactor NOR (PCR-003). The hold rule judges what the hold adds, and the hold does not lower acidic variants by more than the difference between the day 0 aliquots.

Level entering the hold, for the upper limit the highest and for the lower limit the lowest level expected in the pool in manufacture, the upper or lower end of the 95 % prediction interval of a single run over the NOR. It comes from the production bioreactor model for aggregate and acidic variants, and from the fragment model of this step for fragments. Change over the hold, in percentage points, for the upper limit the largest increase and for the lower limit the largest fall from day 0 at the same temperature (0.0 where none), from the results as the hold stability table shows them. Level after the hold, the level entering the hold plus the change.

For aggregate, acidic variants and fragments the largest increase at each temperature, added to the highest level expected in the pool, stays within the criterion of the low-pH inactivation load. On this evidence a hold of 7 days at elution pH is supported at both 2-8 °C and 21 °C for an eluate made at set-point (finding F9). An eluate made at the low end of the elution buffer pH NOR, or one that carries more host cell protein, was not held. At a lower pH, aggregation and hinge hydrolysis are expected to be faster, and more host cell protein could bring more protease activity that clips the antibody. The results therefore do not show that the hold is supported for such an eluate.

The hold does not lower acidic variants. Their largest fall at any sample point was 0.1 points, no more than the 0.6 points between the two day 0 aliquots. The lower limit of 18 % is not met before the hold, at the least favourable corner of the production bioreactor NOR, with culture pH 6.95, culture temperature 34.5 °C, culture duration 18 days and CO2 partial pressure (pCO2) 130 mmHg. There the predicted median of acidic variants is 18.4 %, and the lower end of the 95 % prediction interval of a single batch is 16.2 %, below the limit. This is a result of the production bioreactor, not of the hold. The production bioreactor report PCR-003 accepts it as the producer’s risk and reduces it with alarm limits on culture pH, CO2 partial pressure (pCO2) and osmolality inside the NOR on the side that lowers acidic variants, and with monitoring of acidic variants at harvest.

5.8 Mechanistic interpretation

Pool host cell protein rises as elution buffer pH falls. A lower pH desorbs more of the host cell protein held weakly on the resin backbone and on the ligand, and this protein leaves the column with the product. On the log scale the pH effect is linear, so each step down in pH multiplies pool host cell protein by the same factor.

Pool host cell protein also rises with protein load, as in the A-Mab case study (CMC Biotech Working Group 2009). PCP-005 assumed no mechanism for this rise (PCP-005 §4.1), and the designs do not test one. One explanation would give effects that multiply. Suppose a higher load leaves more host cell protein held weakly on the resin for each gram of antibody bound, and a lower elution pH releases a larger share of what is held. The host cell protein in the pool per mg of antibody is then the product of the two. The designs do not distinguish this explanation from others.

End of pool collect lowers pool host cell protein per mass of antibody. The weakly held impurity desorbs early, at the elution front, so collecting further into the tail of the peak adds antibody with little host cell protein. This does not explain why the effect had the opposite sign in the screening design (finding F11).

Pool host cell protein is slightly higher at both ends of the flow range than at the set-point. A short residence time at high flow lowers the dynamic binding capacity and broadens the mass transfer zone, which could raise pool host cell protein, but the same mechanism predicts no rise at low flow. The rise is at most 14 % across the range studied and 4 % within the NOR, and it is not interpreted further.

Step yield behaves as Section 2 expected. Load flow rate costs yield only at high load, where the shorter residence time lets antibody break through near the capacity of the bed. The gain from extending the end of pool collect becomes smaller as the tail of the elution peak is exhausted, which is the negative square of the model.

Leached Protein A shows no dependence on the parameters within the range studied. Neither of the two opposite effects of protein load expected in Section 2, a longer exposure of the ligand to the harvest proteases and a dilution of the leached ligand over more antibody, is resolved against the variation of the runs. The offset of leached Protein A between the scales is expected to come from the protease activity of the held load, which is not a setting of a run (Section 3.1).

Fragments by nrCE-SDS fall slightly as elution pH rises, which is the direction expected from acid-catalysed hinge hydrolysis during elution. The effect is small because the antibody spends only the duration of the elution at low pH in the column.

For operation, the risk to pool host cell protein lies at high protein load combined with low elution buffer pH, and never with every parameter of the multivariate design inside its NOR (Section 6). Within their NORs, end of pool collect moves pool host cell protein by 8 % and load flow rate by at most 4 %.

6 Design space

The design space of this step is the region in protein load and elution buffer pH, the 2 WC-CPPs, over which a single batch is predicted to meet every intermediate acceptance criterion of the step. A point lies inside it when both ends of the 95 % prediction interval of a single run lie within the criterion of each quality attribute, at manufacturing scale where an offset applies. The criteria of this step are upper limits, so the upper prediction limit of Section 3.5 decides, and at the boundary of the region a single batch is predicted to exceed the criterion with a probability of 2.5 %. Leached Protein A and fragments meet their criteria on this statistic everywhere in the range studied, and step yield has no criterion, so pool host cell protein alone bounds the region. The fractions below are of a grid of 41 points per factor over the range studied, with the offset of pool host cell protein at its estimate.

With load flow rate and end of pool collect at their set-points, the upper prediction limit meets the criterion over 91.0 % of the plane of protein load and elution buffer pH studied. It exceeds the criterion in the corner of high load and low pH, above a protein load of 30.9 g/L resin at pH 3.2 and below pH 3.45 at 50 g/L resin. The excluded corner reaches into the protein load NOR and into the pH NOR, but never into both at once. It begins below pH 3.32 at the upper edge of the protein load NOR, 40 g/L resin, and above 46.3 g/L resin at the lower edge of the pH NOR, 3.4. The upper prediction limit is 25,690 ng/mg at the corner of high load and low pH of the range studied, and 14,518 ng/mg at that corner of the two NORs. With load flow rate and end of pool collect at the corners of their NORs, between 87.5 and 92.1 % of the plane meets the criterion, and at the corners of their ranges studied between 80.2 and 91.0 %.

Over the whole range studied of all 4 factors, 89.3 % of the grid lies inside the design space. The excluded points lie at a protein load of 22 g/L resin or more with elution buffer pH 3.57 or less. No point is excluded with every factor inside its NOR. With load flow rate and end of pool collect within their NORs, each excluded point lies outside the protein load NOR of 20–40 g/L resin or outside the pH NOR of 3.4–3.7. Inside both the protein load NOR and the pH NOR, a point is excluded only where load flow rate is outside its NOR. Most such points lie at a load flow rate of at most 125 cm/hr with end of pool collect at most 2.36 CV. The others lie at 300 cm/hr, the upper end of the flow range studied, at a protein load of 40 g/L resin and end of pool collect at 2 CV, the lower end of its range studied.

Drawn on the median prediction, for comparison only, the region would cover 97.0 % of the grid. A median can meet the criterion where a single batch does not.

Over the NOR the selected model predicts a median pool host cell protein of 4,259 to 12,474 ng/mg at manufacturing scale (Table 7.2). The upper prediction limit reaches

at most 15,697 ng/mg over the NOR, below the criterion. Movement outside the NOR and within a PAR would be assessed under change control.

7 Proven acceptable ranges

7.1 Acceptance basis

Each output of the step is judged against the intermediate acceptance criterion that PCMP-001 §7.3 derives and that PCP-005 §8.1 applies to this step. The derivation is a priori, which here means that it uses the drug substance criteria and the platform worst cases of X-Mab, Y-Mab and Z-Mab, and no data from this campaign. The drug substance criterion is carried backward through the train with the worst case that each downstream step showed on the platform, and no margin is applied. PCMP-001 Version 2.0, which states this rule and the one below, took effect on 2026-09-30, after every study of this report had ended.

PCMP-001 §7.5 then checks each platform worst case against the A-Mab results of its step. The check uses the one-sided 95 % prediction bound of a single batch, at the least favourable combination of NOR low, set-point and NOR high of the response-surface factors of the downstream step, on its unfavourable side. These are the NORs as characterized. The scale-down prediction of the downstream step is first taken to manufacturing scale with the offset of that step at its estimate, as PCMP-001 §7.1 rule 2 requires, so the check is made at manufacturing scale. Where the worst case that the model predicts within the NOR is worse than the platform value, the bound replaces the platform value and the criteria upstream become stricter. Where the bound is better than the platform value, and the model behind it meets the first three adequacy criteria of PCMP-001 §7.2 (a significant model, no significant lack of fit and the limit on overfitting) and is not an exact fit, the bound replaces the platform value and the criteria upstream become wider. The fourth criterion, on the residuals and on influential runs, does not decide the change, and the report that gives the bound examines whether the bound depends on a single run. Otherwise the platform value stands. A point prediction never replaces it.

The bound is that of a single batch, not a confidence bound of the mean, for three reasons that PCMP-001 §7.5 gives. A quality attribute is judged on a single batch, and a single batch can fall beyond a bound of the mean. The platform worst case that the bound may replace was itself the result of a single run. A bound from a tolerance interval, which covers a stated share of all batches, would be statistically purer, but too conservative to set against a platform worst case of one run. Table 7.1 gives the downstream values on which the criteria of this step rely.

Table 7.1: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign			Outcome
				At the edges	Within the NOR	95 % bound	
Low-pH viral inactivation	Aggregates, HMW	Increase, largest (points)	0.70	0.82	0.50	0.54	Better, widened[a)]
	Fragments, LMW		0.15	0.22	0.10	0.19	Confirmed
	Acidic variants		1.00	0.33	0.13	0.13	Confirmed, not widened[b)]
Cation exchange	Host cell protein	Clearance factor, lowest	40.0	27.2	40.2	32.9	Confirmed
	Aggregates, HMW		2.00	1.00	1.58	1.40	Not confirmed, tightened[c)]
	Fragments, LMW		1.30	1.28	1.45	1.27	Confirmed
	Leached Protein A		3.16	not a response[d)]			
Anion exchange	Host cell protein	Clearance factor, lowest	4.00	3.55	5.40	4.18	Better, widened[a)]
	Leached Protein A		10.0	not a response[d)]			
UF/DF	Aggregates, HMW	Increase, largest (points)	0.00	0.07	0.06	0.10	Not confirmed, tightened[c)]

[a)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case. The criteria upstream of this step were recalculated with the bound.
[b)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case, but the model is not adequate by the criteria of PCMP-001: it fails a test, or it fits the results exactly and the tests cannot be applied. The platform value stands.
[c)] The criteria upstream of this step were recalculated with the one-sided 95 % prediction bound of a single batch at the value predicted within the NOR.
[d)] Not a response of the response-surface study at this step, so the platform value stands unverified.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

Each of the 10 rows bears on this report. Host cell protein at cation exchange and anion exchange, fragments at low-pH viral inactivation and cation exchange, and leached Protein A at cation exchange and anion exchange set the criteria of the Protein A pool. Aggregate at low-pH viral inactivation, cation exchange and UF/DF, and acidic variants at low-pH viral inactivation, set the criteria of the held eluate (Section 4.5). For acidic variants the single-batch bound of the low-pH increase is better than the platform value but does not replace it. The acidic variant results of the low-pH design have no scatter, so their model fits them exactly and cannot be judged adequate.

For pool host cell protein the drug substance limit of ≤ 100 ng/mg is carried back through cation exchange and anion exchange. With the platform worst cases of 40-fold and 4-fold clearance, the a priori criterion was ≤ 16,000 ng/mg.

At cation exchange the platform worst case for host cell protein clearance stands. The cation exchange model, from the 28 runs of its response-surface design with 4 centre points, predicts a lowest host cell protein clearance within its NOR of 40.2-fold, no

worse than the platform value of 40-fold, so no criterion is tightened. Its single-batch bound, 32.9-fold, is not better than the platform value, so none is widened.

At anion exchange the single-batch bound for host cell protein clearance replaced the platform worst case. The anion exchange model predicts a lowest clearance within its NOR of 5.40-fold, and its single-batch bound, 4.18-fold, is better than the platform value of 4-fold. The model behind the bound meets the first three adequacy criteria of PCMP-001 §7.2. Its F-test is significant ($p < 0.001$), its lack of fit test is not ($p = 0.824$), and its predicted R^2 falls short of R^2 by 0.02, within the limit of 0.20. The model is not an exact fit, and it meets the fourth criterion, because no run has a Cook's distance above the median of $F(p, n - p)$, 0.81. Runs 2 and 9 lie above 4/n and were examined. With each of runs 2 and 9 deleted in turn, and the model selection and the bound computed again, the selected model still met the three criteria and the bound stayed better than the platform value, at 4.24 to 4.47-fold (INV-002 §8.4). The bound therefore does not depend on a single run. The bound comes from the 28 runs of the planned anion exchange design, with 4 centre points. The criteria of this report were fixed when INV-001 closed on 2026-09-18, before the 12 runs that INV-002 added to that design were complete (study S-AEX-AUG, 2026-09-10 to 2026-09-24). On all 40 runs the bound is the same to three significant figures, 4.18-fold. For these reasons the clearance of anion exchange for A-Mab is known well enough to replace the platform value. The 40-fold of cation exchange and the 4.18-fold of anion exchange give a combined clearance of 167-fold and the criterion in force of ≤ 16,719 ng/mg, 4 % wider than the a priori value. At the edges of their ranges studied, outside their NORs, the lowest clearance measured in a run was 27.2-fold at cation exchange and 3.55-fold at anion exchange, both below their platform values. The criterion therefore relies on both steps running within their NORs (PCR-007, PCR-008).

The widening of the criterion for pool host cell protein is argued by its risk, in the four parts that PCMP-001 §7.5 asks for.

1. The risk is the producer's. A wider criterion admits a Protein A pool with more host cell protein, and a drug substance batch that then exceeded its criterion would be rejected at release.
2. An excursion would be detected at release of the drug substance, where host cell protein is tested by ELISA (AMV-3012). The limit of quantitation of the ELISA, 1 ng/mg, is far below the criterion of ≤ 100 ng/mg. The coverage of the ELISA, the share of the host cell proteins that its antibodies recognise, is 80 %, measured by 2D western blot against the platform CHO host cell protein antigen (AMV-3012). A host cell protein outside that share would not be detected by the release test.
3. A pool at the criterion meets the drug substance limit only if cation exchange clears at least 40-fold, to the cation exchange pool criterion of ≤ 418 ng/mg, and anion exchange at least 4.18-fold. Cation exchange is credited with its platform value, which lies above its own single-batch bound of 32.9-fold. If both steps cleared only their single-batch bounds, a pool at the criterion would give 121 ng/mg in the drug substance, above the limit. Over the NOR the upper prediction limit of a single pool is 15,697 ng/mg, below the criterion.
4. In the integrated process model no simulated batch of 2,000 exceeded the host cell protein criterion of the drug substance, and the normal tail probability is

below one per million (Section 8).

The widening of the criterion for pool host cell protein rests on the model selection of INV-002 corrective action CA3. Without the model selection, the full quadratic model of anion exchange fails the limit on overfitting of PCMP-001 §7.2, with predicted R^2 0.27 below R^2, so no bound replaces a platform value, and the criterion would be the a priori value of ≤ 16,000 ng/mg. Against that criterion, as against the criterion in force, the scale-down model would not be qualified for pool host cell protein (Section 3.1). The NOR-propagated PAR of elution buffer pH for this output would be 3.25–3.9 instead of 3.22–3.9. No conclusion over the NOR would change.

For fragments the drug substance limit of ≤ 2.00 % is carried back through cation exchange and then through the largest increase of low-pH viral inactivation, which gives ≤ 2.45 %. Both point predictions confirm the platform values. Cation exchange is predicted to clear at least 1.45-fold within its NOR against 1.30-fold, and low-pH viral inactivation to add at most 0.095 points against 0.15. Neither single-batch bound is better than its platform value, at 1.27-fold and 0.194 points, so both platform values stand and the criterion in force is the a priori value. Both fragment models also fail the limit of PCMP-001 §7.2 on how far predicted R^2 may fall below R^2, by 0.21 and 0.33 against 0.20.

For leached Protein A the drug substance limit of ≤ 5 ppm is carried back through cation exchange and anion exchange, which together remove it 31.6-fold on the platform. That gives ≤ 158 ppm. Leached Protein A is not a response of the cation exchange or anion exchange designs. Its platform values therefore cannot be checked in this campaign, and they stand. No criterion upstream of this step uses a platform worst case of the Protein A step.

None of the 3 criteria of the Protein A pool was tightened. The aggregate criterion of the held eluate, the low-pH inactivation load, which the A-Mab results of cation exchange and low-pH viral inactivation changed, is derived in Section 4.5.

7.2 Outputs against their criteria over the NOR

Table 7.2 gives each output at the set-point and over the NOR, as predicted by its selected model, and the 95 % limits of a single run over the NOR, which decide the verdict. Pool host cell protein and leached Protein A are at manufacturing scale. Fragments are in the scale-down model, because they carry no offset.

Table 7.2: Protein A outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model[a)]		95 % limits of a single run over the NOR[a)]	Met across the NOR[b)]
		At the set-point	Range over the NOR		
Host cell protein (ng/mg)	≤ 16,719	7,150[c)]	4,259 to 12,474[c)]	3,374 to 15,697[c)]	Yes
Leached Protein A (ppm)	≤ 158	4.12	4.12 to 4.12	2.72 to 5.52	Yes
Fragments, LMW (%)	≤ 2.45	0.92	0.88 to 0.96	0.75 to 1.09	Yes

[a)] At manufacturing scale: the scale-down prediction minus the offset of the scale-down model for this output, or divided by it where the offset is a ratio.
[b)] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty).
[c)] Modelled on the log scale: the median, the back-transformed mean prediction.
Fragments carry no offset, and their values are those of the scale-down model.

The 3 outputs with a criterion all meet it at the set-point and across the NOR in force. Each verdict rests on the upper end of those limits, the upper prediction limit of a single run over the NOR. It is 15,697 ng/mg for pool host cell protein against ≤ 16,719 ng/mg, 5.52 ppm for leached Protein A against ≤ 158 ppm, and 1.09 % for fragments against ≤ 2.45 %. The limit for pool host cell protein is taken with the offset ratio at its estimate. With the ratio at the low end of its interval it is 17,143 ng/mg, above the criterion, and for this reason the scale-down model is not qualified for pool host cell protein over the NOR in force (Section 3.1). The table gives medians for pool host cell protein, because its model is on the log scale. The model of leached Protein A has no term, so its prediction is the same at every setting.

7.3 Proven acceptable ranges per attribute and parameter

Table 7.3 gives the PAR of each parameter of the multivariate design for each output with a criterion, by the univariate analysis and by the NOR-propagated analysis of PCMP-001 §6.4. Every PAR spans the range studied of its parameter, except the NOR-propagated PAR of elution buffer pH for pool host cell protein. That PAR is 3.22–3.9. Its lower end lies 0.017 pH units inside the range studied, two steps of the grid of the analysis. The lower edge of the pH NOR lies 0.18 pH units above the end of this PAR, and the set-point 0.33.

Table 7.3: Proven acceptable ranges for each output with a criterion and each parameter of the multivariate design.

Parameter	Range studied	HCP ≤ 16,719 ng/mg		LPA ≤ 158 ppm		Fragments ≤ 2.45 %	
		Univ.	NOR-prop.	Univ.	NOR-prop.	Univ.	NOR-prop.
Protein load (g/L resin)	10–50	10–50	10–50	10–50	10–50	10–50	10–50
Elution buffer pH (pH)	3.2–3.9	3.2–3.9	3.22–3.9	3.2–3.9	3.2–3.9	3.2–3.9	3.2–3.9
Load flow rate (cm/hr)	100–300	100–300	100–300	100–300	100–300	100–300	100–300
End of pool collect (CV)	2–3.2	2–3.2	2–3.2	2–3.2	2–3.2	2–3.2	2–3.2

Univ., the univariate PAR, with the other parameters at their set-points, judged on the median prediction for pool HCP and on the mean prediction for the other outputs. NOR-prop., the NOR-propagated PAR, a multivariate PAR with the other parameters varied within their NORs, judged on the upper end of the 95 % prediction interval of a single run. Each PAR is judged on the selected model of the output. Pool HCP and LPA are judged at manufacturing scale and fragments in the scale-down model, which carries no offset for them. HCP, pool host cell protein. LPA, leached Protein A.

A PAR that reaches the edge of the range studied is reported as reaching it, and no claim is made beyond it (PCMP-001 §7.6). Each NOR-propagated PAR holds while the other parameters stay within their NORs. It does not cover two parameters at the edges of their ranges together. Towards the corner of high protein load and low elution buffer pH, beyond the NOR of at least one of the two, the upper prediction limit of pool host cell protein exceeds its criterion even with load flow rate and end of pool collect at their set-points (Section 6).

Figure 7.1 shows the PAR of elution buffer pH for pool host cell protein, the parameter with the largest main effect on it, and Figure 7.2 the PAR of elution buffer pH for fragments. In each figure the left panel shows the univariate analysis and the right panel the NOR-propagated analysis, with the acceptable region shaded green. Pool host cell protein is shown at manufacturing scale and fragments in the scale-down model. No PAR figure is shown for leached Protein A, because its selected model has no term and therefore no parameter with a largest effect.

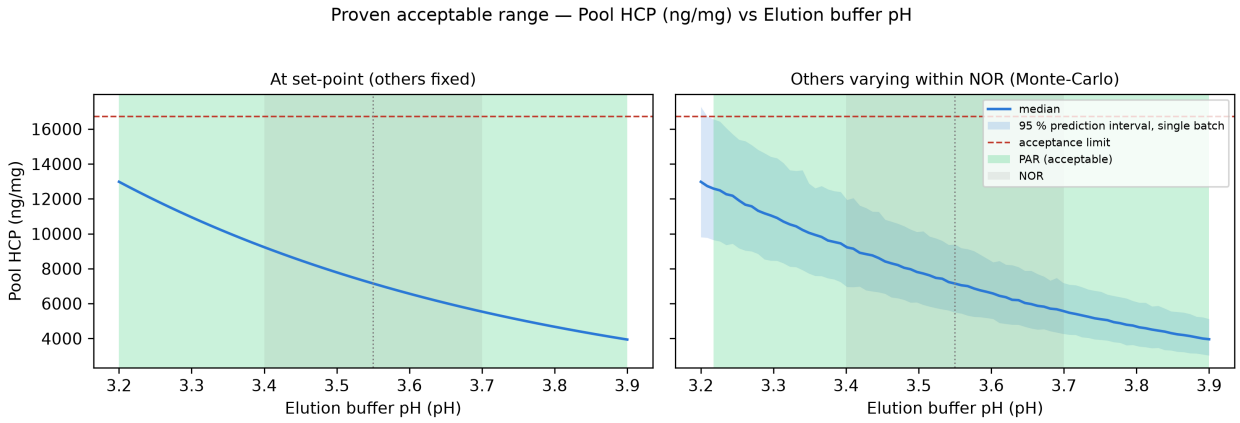


Figure 7.1: PAR of elution buffer pH for pool HCP (ng/mg), at manufacturing scale. Left, univariate analysis. Right, NOR-propagated analysis (95 % prediction interval of a single run). Green, the region that meets the criterion.

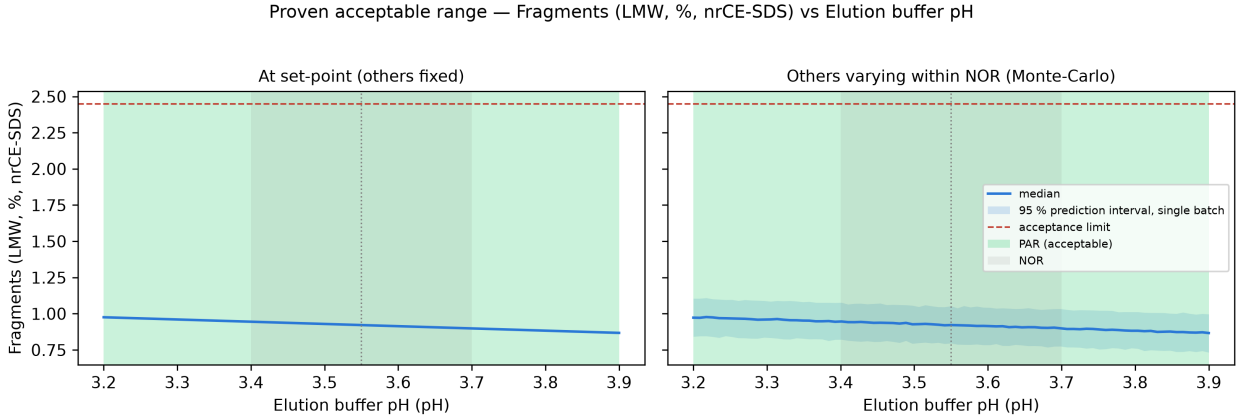


Figure 7.2: PAR of elution buffer pH for fragments by nrCE-SDS (%), in the scale-down model. Left, univariate analysis. Right, NOR-propagated analysis (95 % prediction interval of a single run). Green, the region that meets the criterion.

Operating temperature and bed height have no model-based PAR. For pool host cell protein, step yield and leached Protein A their ranges rest on the measured responses at each end of the range studied, with the other parameters at their set-points (Section 5.6).

8 Process capability and robustness

Table 8.1 gives the capability at commercial scale, from the integrated process model of the train (Section 3.5), of the drug substance attributes to which this step contributes.

Table 8.1: Commercial-scale capability of the drug substance attributes to which the Protein A step contributes.

CQA	Crit.	Criterion	Mean	Range	Limiting side	Cpk
Host Cell Protein (HCP)	M-H	≤ 100 ng/mg	15.2	5.54–42.3	upper	6.14
Leached Protein A	L-M	≤ 5 ppm	< LOQ[a)]	< LOQ[a)]	upper	n.m.[b)]
Fragments (LMW, clipped species)	L	≤ 2 %	0.643	0.512–0.824	upper	10.62
Residual DNA	VL	≤ 0.001 ng/dose	0.0002	0.0001–0.0004	upper	7.30
Acidic charge variants (deamidation)	VL	18–40 %	22.4	17.3–27.3	lower	1.05

[a)] Below the limit of quantitation (LOQ) of the ELISA, 0.25 ppm (AMV-3016). [b)] Not meaningful, because the simulated values lie below the LOQ and cannot be measured. Crit., criticality. Mean and range of the 2,000 simulated batches. Limiting side, the limit of the criterion nearer to the simulated batches.

Host cell protein and fragments have Cpk values of 6.14 and 10.62, above the target of 1.33 of PCMP-001 §7.9.

Residual DNA has no intermediate acceptance criterion. It is carried through the train in pg per mg of antibody and judged in the drug substance in ng per dose. The dose is the 10 mg/kg of the A-Mab case study (CMC Biotech Working Group 2009) for an adult of 70 kg, 700 mg of antibody. The body weight is an assumption of this report. In the simulation the drug substance carries at most 0.0004 ng per dose against the criterion of ≤ 0.001 ng per dose, a Cpk of 7.30. The integrated process model carries a nominal DNA reduction for each chromatography step, 2.5 log10 at Protein A, 1.5 at cation exchange and 5 at anion exchange. The platform history gives no worst-case DNA clearance for these steps (Section 2), so the Cpk of residual DNA rests on these nominal values. At this step the reduction measured at the centre points, 2.2 to 2.5 log10, is at or below the nominal value. At anion exchange every pool of the response-surface design was below the limit of quantitation of the qPCR, 1 pg/mg (AMV-3014), so the reduction there is not measured. Each run shows only that it exceeds the DNA in its load over that limit, between 1.2 and 1.5 log10 (PCR-008).

The Cpk of residual DNA therefore rests on a nominal anion exchange reduction that the data of this campaign do not measure.

No simulated batch of 2,000 fell outside the drug substance criterion for host cell protein, leached Protein A, fragments or residual DNA. With none of 2,000 outside, the probability of an out-of-specification batch in the simulation is below 0.15 % at 95 % confidence. The bound does not include the uncertainty of the integrated process model itself.

The Cpk of leached Protein A is not meaningful. The simulated drug substance values average 0.003 ppm, below the limit of quantitation of the ELISA (0.25 ppm, AMV-3016). They cannot be measured, and their spread is that of the model, not of a result. Leached Protein A is judged instead against its criteria in the pool (Section 7) and in the drug substance.

Acidic variants have the lowest Cpk of the drug substance, 1.05, set by the lower limit of 18 %. This is below the target of 1.33 and above the minimum of 1.00 of PCMP-001 §7.9, which asks PCMR-001 to state the parameters that drive it. In the simulation 1 batch of 2,000 fell outside the criterion, and the normal tail probability is 821 per million. The Protein A step does not separate charge variants. The hold of the eluate does not lower acidic variants (Section 5.7), so it does not move them towards the lower limit that sets the Cpk.

9 Parameter classification

Parameters were classified under SOP-4001 by the decision logic of PCMP-001 §7.7, as PCP-005 §7 applies it (criterion 7). An attribute is affected by a parameter when a term of the selected model that contains the parameter is significant at α = 0.05 (PCMP-001 §6.2). A parameter whose effect on a CQA is significant and practically relevant is critical. A critical parameter is a WC-CPP when it has a low risk of leaving its acceptable range in operation, because it is readily controlled and its PAR is wide against its NOR, and a CPP otherwise. A KPP affects process performance and has no relevant effect on a CQA, and a GPP has no relevant effect on either. A significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical. Neither PCMP-001 nor SOP-4001 sets a numerical threshold for a practically relevant effect or for a low risk. Each class below is therefore argued from the size of each effect across the PAR, stated beside its significance, and from the margin of the PAR to the NOR. The effect on pool host cell protein is given at manufacturing scale with the other parameters at their set-points, and as a share of its criterion of ≤ 16,719 ng/mg.

- Protein load was classified as a WC-CPP. Its terms are significant for pool host cell protein, A ($p < 0.001$), and for step yield, A ($p < 0.001$) and AC ($p = 0.027$), but not for fragments ($p = 0.079$). Across its PAR for pool host cell protein, which is its range studied of 10–50 g/L resin, pool host cell protein rises from 4,590 to 11,137 ng/mg, a change of 39 % of the criterion. Together with elution buffer pH it takes pool host cell protein above its criterion within the range studied, and 8 results of the designs at the corner of high load and low pH exceeded the criterion (Section 5.3.1). The effect is therefore practically relevant, and protein load is critical. The load is set from the load volume and the titre of the clarified harvest by Protein A affinity HPLC (AMV-3021), so it is readily controlled, and its PAR extends 10 g/L resin below and 10 g/L resin above its NOR of 20–40 g/L resin. Over that NOR the scale-down model is not qualified for pool host cell protein, and the NOR is flagged for discussion with manufacturing (Section 3.1).
- Elution buffer pH was classified as a WC-CPP. Its terms are significant for pool host cell protein, B ($p < 0.001$), and for fragments, B ($p = 0.001$), and step yield has no term in it. Across its PAR for pool host cell protein, 3.22–3.9, pool host cell protein falls from 12,598 to 3,939 ng/mg as the pH rises, a change of 52 % of the criterion. Fragments fall by 0.11 percentage points across their PAR, the range studied, 4 % of their criterion. The effect on pool host cell protein is practically relevant, and with protein load it bounds the design space. Buffer pH is controlled at preparation and verified at release, and the PAR for pool host cell protein extends 0.18 pH units below and 0.20 above the NOR of 3.4–3.7. The study event DEV-005-01 showed that a buffer below the NOR can pass release

when the pH meter is incorrectly calibrated (Section 13.1). The buffer of that event, at pH 3.38, lay below the NOR and inside the PAR.

- Load flow rate was classified as a KPP. Its terms are significant for step yield, C ($p = 0.023$) and AC ($p = 0.027$), and for pool host cell protein, C^2 ($p = 0.025$). It costs step yield at high load, up to 3.6 percentage points across its range studied at 50 g/L resin, because a faster load shortens the residence time and lowers the dynamic binding capacity. Its effect on pool host cell protein is the curvature, a change of at most 979 ng/mg across its PAR, the range studied of 100–300 cm/hr, or 6 % of the criterion, and of at most 4 % within the NOR. That effect is significant but too small to move pool host cell protein appreciably within the PAR, so it does not make load flow rate critical. The PAR extends 50 cm/hr below and 50 cm/hr above the NOR of 150–250 cm/hr. No effect of load flow rate on leached Protein A or fragments was detected within the range studied.
- End of pool collect was classified as a KPP. Its terms are significant for step yield, D ($p < 0.001$) and D^2 ($p = 0.035$), and for pool host cell protein, D ($p = 0.003$). It has the largest effect on step yield, a gain of 7.3 percentage points across its range studied. Across its PAR for pool host cell protein, the range studied of 2–3.2 CV, it lowers pool host cell protein from 7,761 to 6,587 ng/mg, a change of 7 % of the criterion, and the sign of that effect differs between the designs (finding F11). That effect is significant but too small to move pool host cell protein appreciably within the PAR, so it does not make end of pool collect critical. The PAR extends 0.3 CV below and 0.3 CV above the NOR of 2.3–2.9 CV. The class would be revisited if a later study confirmed the screening sign.
- Operating temperature was classified as a GPP. No difference from the set-point runs was detected in pool host cell protein, step yield or leached Protein A at either end of its range. The intervals allow a rise in pool host cell protein of up to 4,291 ng/mg. Such a rise would still leave pool host cell protein within its criterion at manufacturing scale, and every individual result met its criterion (Section 5.6). The range studied of 15–30 °C, which these results support, extends 3 °C below and 4 °C above the NOR of 18–26 °C. The class rests on the attributes measured. Fragments were not measured by nrCE-SDS in the univariate study (Section 5.6), so for fragments the class rests on the mechanism and on the response-surface model. A higher temperature is expected to speed acid-catalysed hinge hydrolysis during elution. In the response-surface design fragments changed by 0.11 percentage points across the whole range of elution buffer pH, the condition that drives that hydrolysis (Section 5.3.4). Their upper prediction limit over the NOR lies 1.36 points below the criterion. The class assumes that a change of temperature within its range studied does not use up that margin, and this was not tested.
- Bed height was classified as a GPP. No difference from the set-point runs was detected in pool host cell protein, step yield or leached Protein A at either end of its range. The intervals allow a rise in pool host cell protein of up to 2,815 ng/mg, and every individual result met its criterion (Section 5.6). The range studied of 10–30 cm, which these results support, extends 5 cm below and 5 cm above the NOR of 15–25 cm. Bed height is fixed by the packing specification of the commercial column and does not vary in operation. A short bed at the upper end of the load range was not characterized, and no claim is made for it. The class rests on the attributes measured. Fragments were not measured by

nrCE-SDS in the univariate study (Section 5.6), so for fragments the class rests on the mechanism and on the response-surface model. A fragment that carries Fc binds the ligand as the intact antibody does, so a shorter residence time is expected to lower the capture of both alike. In the response-surface design no effect of load flow rate on fragments was detected over residence times from 4 to 12 min, at the set-point bed height (Section 5.3.4). The shortest bed gives 3 min, below that span, and no fragment result supports the class there.

In summary the step has 2 WC-CPPs, 2 KPPs and 2 GPPs. No parameter is in the class CPP. The two WC-CPPs are critical parameters that are well controlled. The classes of protein load and elution buffer pH rest on the size of their effects on pool host cell protein within the PAR and on the corner where the two together take it above its criterion. Both effects agree in direction with the A-Mab case study. Load flow rate and end of pool collect also have significant effects on pool host cell protein, each too small to make the parameter critical. The KPP classes rest on the model for step yield, on the direction of the yield effects in both designs, and on the mechanism of breakthrough and of peak collection.

10 Contribution to the control strategy

The Protein A step contributes to the control strategy in the following ways.

1. Process parameters. Protein load and elution buffer pH are controlled within their NORs as WC-CPPs, and load flow rate and end of pool collect within their NORs as KPPs. The NORs are those of PCP-005. Over the NOR of protein load the scale-down model is not qualified for pool host cell protein, and that NOR is flagged for discussion with manufacturing on whether it can be improved (Section 3.1). Movement outside the NOR and within the PARs of Section 7 is assessed under change control.
2. Material controls. Elution buffer is released against its pH specification, on a calibrated pH meter (DEV-005-01).
3. Resin and load. The protein load is set from the titre of the clarified harvest by Protein A affinity HPLC (AMV-3021). Leached Protein A is highest on new resin and is expected to rise with the protease activity of the load. Resin use, cleaning and sanitization are controlled by SOP-2008. The hold of the clarified harvest before capture, the expected cause of the offset of leached Protein A between the scales, is characterized in PCR-004.
4. Hold. An eluate made at set-point may be held at elution pH for up to 7 days at 2-8 °C or 21 °C before viral inactivation. An eluate made at other settings within the NOR was not held (Section 5.7).
5. Downstream clearance. Host cell protein, leached Protein A and fragments are further cleared by cation exchange (PCR-007) and anion exchange (PCR-008). The criterion for pool host cell protein relies on cation exchange clearing host cell protein at least 40-fold, its platform worst case, and anion exchange at least 4.18-fold, its single-batch bound, while both run within their NORs (Section 7). Host cell protein is tested at release of the drug substance. Viral clearance is claimed at low-pH viral inactivation (PCR-006), anion exchange and virus filtration (PCR-009), and not at this step.

The control strategy of the whole train is consolidated in PCMR-001.

11 Discussion

The characterization confirms the main effects expected for this step. Elution buffer pH and protein load have the largest effects on pool host cell protein, and on the log scale their effects multiply. Whether a further interaction of the two exists the data do not decide (Section 5.3.1). End of pool collect, protein load and load flow rate have the largest effects on step yield. Leached Protein A shows no dependence on the parameters within the range studied, and fragments depend only slightly on elution buffer pH. End of pool collect lowers pool host cell protein in the response-surface design but raised it, not significantly, in the screening design, for a reason that was not identified.

Model selection gave leached Protein A and fragments models that meet the adequacy criteria on lack of fit and predictive power, which their full quadratic models do not. The choice of model is a discrepancy from the plan, so its effect on the conclusions was examined. The conclusion over the NOR is the same with each of the 9 models within 2 AICc units of the selected model of pool host cell protein, and with run 23 analysed in each of three ways.

Over the NOR in force the scale-down model is not qualified for pool host cell protein. The conclusion holds with the offset ratio at its estimate and fails at the low end of its interval, at the upper end of the protein load NOR. This does not depend on the criterion, on the 6 set-point runs added after the planned runs had been analysed, or on the choice of model (Section 3.1). The NOR is not narrowed, because NORs come from production. It is flagged for discussion with manufacturing on whether it can be improved, and the limit at the low end of the ratio meets the criterion when the upper edge of the protein load NOR is 38.9 g/L resin. The offsets were estimated at set-point only. The report therefore assumes that the ratio for pool host cell protein and the difference for leached Protein A are the same across the ranges studied, and this assumption is not tested. Fragments have no measured offset. Their conclusions assume no difference between the scales, on the mechanism of capture and on the bound from the hold study of the clarified harvest (Section 3.1).

The criterion for pool host cell protein links this step to the two later steps. It credits cation exchange with its platform worst case of 40-fold, above the single-batch bound of 32.9-fold of its A-Mab model, and anion exchange with its single-batch bound of 4.18-fold. The control of this step therefore depends on the control of those steps (PCR-007, PCR-008), and an excursion would be detected at release of the drug substance (Section 7).

The models of leached Protein A and fragments fail criterion 5 on a residual trend, and their conclusions rest on its size (PD-005-02). The re-assay of run 11 decides whether leached Protein A depends on end of pool collect, but not its judgement against the criterion (Section 13.2). Each design used one feed, so the effect of feed quality was

not estimated. The univariate study cannot detect interactions, has 2 runs at each end and no centre run (PD-005-07), and did not measure fragments by nrCE-SDS. The hold study held one eluate, made at set-point.

12 Conclusions

1. The step was characterized under PCP-005 in a screening design, a response-surface design and a univariate study.
2. The scale-down model is qualified by its offset for leached Protein A. Over the NOR in force, with protein load at 20–40 g/L resin, it is not qualified for pool host cell protein. The conclusion holds with the offset ratio at its estimate and fails at the low end of its interval. The NOR is not narrowed. It is flagged for discussion with manufacturing on whether it can be improved.
3. Each selected model of the response-surface outputs meets the adequacy criteria on lack of fit and predictive power, and each model with a term is significant. Criterion 5 is met except for leached Protein A and fragments, whose residual trends are small against their criteria (PD-005-02).
4. Elution buffer pH and protein load have the largest effects on pool host cell protein, and their main effects multiply. Leached Protein A shows no dependence on the parameters within the range studied, on the re-assay of run 11, and fragments fall slightly as elution buffer pH rises.
5. The criterion for pool host cell protein, ≤ 16,719 ng/mg, is widened by the single-batch bound of anion exchange. The widening is the producer's risk, and an excursion would be detected at release of the drug substance.
6. Each output with a criterion met it across the NOR in force, pool host cell protein with the offset ratio at its estimate. Every NOR-propagated PAR spans the range studied of its parameter, except that of elution buffer pH for pool host cell protein, 3.22–3.9. The design space, drawn on the upper prediction limit of a single run, covers 89.3 % of the range studied and excludes no point with every parameter inside its NOR.
7. Protein load and elution buffer pH are WC-CPPs, load flow rate and end of pool collect are KPPs, and operating temperature and bed height are GPPs. No parameter is in the class CPP.
8. The ranges studied of operating temperature and bed height are supported for pool host cell protein, step yield and leached Protein A, with the other parameters at their set-points.
9. In a hold study of the Version 2.0 re-analysis, what a 7-day hold at elution pH at 2-8 °C or 21 °C added to an eluate made at set-point kept aggregate, acidic variants and fragments within the upper limits of the low-pH inactivation load. The hold did not lower acidic variants, which can enter it below their lower limit (PCR-003).
10. At commercial scale the Cpk of host cell protein and of fragments exceeds the target of 1.33.
11. Neither study event changes the judgement of an output against its criterion. The results roll up into PCMR-001.

13 Study events and discrepancies from the plan

PCP-005 calls a departure from the plan a deviation, but a process characterization study is not GMP manufacture, so this report records the events of execution as study events, under the numbers Version 1.0 gave them, and every departure from the plan as a discrepancy from the plan. Table 13.1 lists the 2 study events of the step. Each was investigated, and its run was retained in the analysis.

Table 13.1: Study events recorded for the studies of the Protein A step.

Event	Summary	Detected during	Outcome
DEV-005-01	Elution-buffer pH prepared below target on an RSM run	post-run review of the buffer preparation record (the buffer had passed release on an incorrectly calibrated pH meter)	run retained in the analysis
DEV-005-02	Leached Protein A result of RSM run 11 initially out of trend; re-assayed	analytical review	run retained in the analysis

13.1 DEV-005-01. Elution buffer pH below target on a response-surface run

The elution buffer of lot LOT-BUF-5290 was used on run 23 of the response-surface design, whose design setting for elution buffer pH was 3.55. During preparation the titration of the buffer overshot its end point, and the buffer was at pH 3.38, 0.17 pH units below its target. It passed release because the pH meter used for release was incorrectly calibrated and read it within its specification. The recorded pH is below the NOR of 3.4–3.7 and inside the range studied of 3.2–3.9.

Run 23 is analysed at its recorded pH, the setting at which it was made. PCP-005 planned every run at its design setting, so this is the discrepancy from the plan PD-005-03 (Section 13). The run gave a pool host cell protein of 8,390 ng/mg at manufacturing scale, below the criterion of ≤ 16,719 ng/mg. On the selected model its residual is −0.21 on the natural log scale. The result is therefore 19 % below the median predicted at pH 3.38, 10,368 ng/mg, which is 2.0 residual standard deviations. Its Cook’s distance, 0.16, is above the limit of 4/n. At the design pH of 3.55 the same model predicts 7,761 ng/mg, closer to the result. The result does not show the full

rise in pool host cell protein that a buffer at the recorded pH would be expected to cause. The data do not establish why.

Table 13.2 gives the conclusion for pool host cell protein with run 23 analysed in three ways, each with the model selection repeated.

Table 13.2: Selected model of pool HCP and its upper prediction limit over the NOR (ng/mg, manufacturing scale) with run 23 analysed in three ways.

Run 23	Scale	Terms	At the estimate	At the low end	Scale-down model qualified
At its recorded pH of 3.38 (used)	log	A, B, C, D, C^2	15,697	17,143	No
At its design pH of 3.55	log	A, B, C, D, AB, A^2, C^2	15,424	16,844	No
Excluded	log	A, B, C, D, AB, A^2, C^2	15,498	16,925	No

Terms of the selected model besides the intercept, coded as in the factor legend. AB, the interaction of A with B. A^2, C^2, the squares of A and C. At the estimate and at the low end, the upper prediction limit with the offset ratio at its estimate and at the low end of its interval. Scale-down model qualified, both limits within the criterion of ≤ 16,719 ng/mg.

With run 23 at its design pH, or excluded, the log scale is chosen again, and the selection adds the interaction of protein load with elution buffer pH and the square of protein load. The interaction is not significant in the full quadratic model on the log scale in any of the three analyses (p = 0.156, 0.069 and 0.081). Nor is it significant in the two selected models that contain it (p = 0.064 and 0.071). In the selected model with the run at its design pH the square of protein load is negative (p = 0.018), so in that analysis the rise of pool host cell protein with load levels off at high load. In all three analyses pool host cell protein meets its criterion of ≤ 16,719 ng/mg at every combination of the NOR with the offset ratio at its estimate, and exceeds it with the ratio at the unfavourable end of its interval. The scale-down model is therefore not qualified for pool host cell protein however run 23 is analysed. The analysis at the recorded pH gives the highest upper prediction limit of the three. The run was retained, and it carries a footnote in Appendix B.

13.2 DEV-005-02. Leached Protein A result of run 11 initially out of trend

The leached Protein A result of run 11 of the response-surface design was out of trend at analytical review. By the rule in the record of the event, a result is out of trend when it lies outside the two-sided 99 % prediction interval of a single result from the full quadratic model of the response-surface design, fitted to the other runs of the design without the result under review, at the settings of the run. The review was made when the design ran, before model selection was adopted on 2026-09-15, so it used the model that PCP-005 specifies. For run 11 the interval, from the other 27

runs with 12 residual degrees of freedom, runs from −1.46 to 5.38 ppm in the scale-down model. Its lower end lies below zero, so at this run only a high result could be out of trend. The first result, 6.37 ppm (7.55 ppm at manufacturing scale), lay above the interval and above every other result of the two designs. The sample was re-assayed under AMV-3016. The re-assay, 2.12 ppm (3.30 ppm at manufacturing scale), lay inside the interval and inside the range of the other runs of the response-surface design, 1.70 to 4.61 ppm. The first result was 3 times the re-assay. The investigation suspected carryover in the ELISA, which would add analyte from a preceding sample, and did not confirm it. The re-assay result is the one reported in Appendix B and used in the analysis, because the investigation attributed the first result to that suspected cause. The first result was kept in the record with its status marked. Both results are far below the criterion of ≤ 158 ppm.

With the first result in place of the re-assay, the analysis changes. The model selection then puts leached Protein A on the log scale and keeps end of pool collect, and that model meets criteria 2 to 4. Its F-test is significant ($p = 0.0495$), its lack of fit test is not ($p = 0.894$), and its predicted R^2 falls short of R^2 by 0.137. On that model leached Protein A would fall by 22 % across the range of end of pool collect. Its upper prediction limit over the NOR would be 6.60 ppm at manufacturing scale, against 5.52 ppm with the re-assay, and every PAR would still span the range studied of its parameter. The trend of the residuals on run order would not be significant ($p = 0.198$). The judgement of leached Protein A against its criterion is therefore the same with either result. The class of end of pool collect would not change, because that effect, 0.74 ppm across its range against the criterion of ≤ 158 ppm, is too small to move leached Protein A appreciably (Section 9). Whether leached Protein A depends on end of pool collect, and whether its residuals drift with run order, depends on which result is used.

13.3 Discrepancies from the plan

The report records 10 discrepancies from the plan, which Table 13.4, Table 13.5 and Table 13.6 list with what PCP-005 planned, what was done and why. They are judged against PCP-005 Version 2.0, the plan this report is written against. That version includes the retest of retained samples, S-PA-RET, made under the method decision AMD-001, so S-PA-RET is not a discrepancy. Four discrepancies change how the runs are analysed or measured and are described where their results are reported. They are the use for the ranges and the design space of the models of leached Protein A and fragments, which do not meet criterion 5 (PD-005-02, Section 5.5), run 23 analysed at its recorded elution buffer pH (PD-005-03, Section 13.1), the model selection in place of the full quadratic model of PCP-005 §5.5 (PD-005-04, Section 3.5), and the loaded antibody mass of step yield taken from the titre by Protein A affinity HPLC in place of absorbance at 280 nm, because absorbance measures every protein in a clarified harvest and not the antibody alone (PD-005-06, Section 3.3). Two are further studies, and three are departures from the plan found in the Version 2.0 review of the record. The sixth is the use of the model of pool host cell protein against a rule of PCMP-001 §7.1. These six are described below.

13.3.1 PD-005-01. Supplementary set-point runs in the scale-down model qualification

PCP-005 planned 3 set-point runs of the scale-down model in the qualification study S-PA-SDM, compared with 5 manufacturing-scale batches. During the Version 2.0 re-analysis, while INV-001 was open, a further 6 set-point runs were made in study S-PA-SDM2 (2026-09-07 to 2026-09-11). Table 13.3 gives the qualification on the 3 planned runs alone.

On the planned runs alone the scale-down model is not qualified for pool host cell protein over the NOR in force. The ratio is 0.935 (0.795 to 1.100) on the planned runs and 0.973 (0.891 to 1.063) on all runs. With the ratio at the low end of the wider interval, the upper prediction limit over the NOR is 19,212 ng/mg, above the criterion of ≤ 16,719 ng/mg, while at the estimate of the ratio the criterion is met. The further runs narrowed the half-width of the interval on the log scale by 46 %. On all runs the limit with the ratio at its low end is 17,143 ng/mg, still above the criterion, so the model is not qualified for pool host cell protein on all runs either (Section 3.1). For leached Protein A the model is qualified on the planned runs as well, with the scale-down model 1.60 ppm lower (0.43 to 2.76 ppm lower).

Table 13.3: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, Protein A, planned runs only.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Host cell protein (ng/mg)			7,400[a)]	7,911[a)]	×0.935[a)]	×0.795 to ×1.100	No / Yes / Yes	No
Leached Protein A (ppm)	3	5	2.10	3.70	−1.60	−2.76 to −0.43	Yes / Yes / Yes	Yes
Step yield (%)			93.6	94.4	−0.8	−4.5 to 2.9	no criterion[b)]	n.a.[b)]
Column efficiency (plates/m)			2,460	2,649	−188	−390 to 13	no criterion[b)]	n.a.[b)]
Peak asymmetry			1.22	1.22	0.00	−0.11 to 0.11	no criterion[b)]	n.a.[b)]

[a)] Modelled on the log scale: geometric means, and the offset as the ratio of the scale-down to the manufacturing geometric mean. The manufacturing-scale value is the scale-down value divided by the ratio.
[b)] The output has no acceptance criterion at this step, so no conclusion depends on its offset and qualification does not apply. The offset is recorded.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

The decision to make the runs was taken during the re-analysis, after the result on the planned runs was known. Runs added after a result was seen raise the chance that a test passes. Here the model is not qualified for pool host cell protein on all runs either, so the added runs did not change the outcome of the qualification. The rule of qualification and the criterion were not changed because of the further runs.

Only the number of runs grew. The centre points of the designs are also set-point runs of the scale-down model. They were not added to the offset, because PCP-005 §5.1 defines the qualification on the runs of the qualification study.

13.3.2 PD-005-05. Hold study of the Protein A eluate

PCP-005 planned no hold study, and Version 1.0 reported no hold result. During the Version 2.0 re-analysis, after INV-001 closed, the eluate was held in study S-PA-HLD (2026-09-19 to 2026-09-28), because Version 2.0 judges a held intermediate against its intermediate acceptance criteria. The plan set no hold criterion, because it foresaw no hold study, so the rule that judges the held eluate against the criteria of the low-pH inactivation load is part of this discrepancy. The study and the rule are described in Section 4.5 and the results in Section 5.7. They support a hold of 7 days at 2-8 °C or 21 °C for an eluate made at set-point.

13.3.3 PD-005-07. Univariate study without runs at the centre

PCP-005 §6.4 requires each parameter of the univariate study to be run at the low edge, the centre and the high edge of its range studied, with the same responses as in the multivariate designs. Study S-PA-UNI (2025-12-08 to 2025-12-12) ran each end of each range 2 times and made no run at the centre. The 3 set-point runs of the qualification study S-PA-SDM (2025-11-05 to 2025-11-14) served as the reference instead. The departure was found in the Version 2.0 review of the study record against PCP-005, and the runs were retained. Without the centre runs the study has no set-point reference of its own. A difference from the reference therefore combines the effect of the parameter with any difference between the two studies. Step yield was higher than the reference at every end of both ranges (Section 5.6), which is consistent with such a difference. The judgement of pool host cell protein and leached Protein A rests on the individual results at both ends, each against its criterion, and does not depend on the reference.

13.3.4 PD-005-08. Leached Protein A offset of the scale-down model accounted for, not investigated

PCP-005 Version 1.0 accepted the scale-down model for an output only where the qualification showed no difference between the scales significant at $\alpha = 0.05$, and it required any difference to be investigated and the model modified before the characterization studies. On the 3 scale-down runs of study S-PA-SDM against the 5 manufacturing-scale batches, leached Protein A differs between the scales under that test ($p = 0.037$, two-sided Student t test with pooled variance). No other output that the qualification compared differs at that level. The difference was not investigated before the characterization studies. The departure was found in the Version 2.0 re-analysis of the qualification against the Version 1.0 rule. PCMP-001 Version 2.0 §7.1 lets a scale-down model differ in mean from manufacturing scale where its offset is known precisely enough, so the Version 1.0 test was stricter than the use of the model

needs. Leached Protein A is judged through its offset, and the model is qualified for it on the planned runs and on all runs (Section 3.1, Section 13.3.1). The expected cause of the difference, the longer hold of the load at manufacturing scale, is stated in Section 3.1 and was not tested.

13.3.5 PD-005-09. Univariate study made before the response-surface design

PCP-005 §11 sets the order of the work. The qualification runs come first, the screening design second and the response-surface design third, and the univariate assessments come fourth and may run in parallel with the response-surface design. Study S-PA-UNI ran from 2025-12-08 to 2025-12-12, after the screening design ended on 2025-12-05 and before the response-surface design began on 2026-01-12 (Table 4.2). The departure was found in the Version 2.0 review of the execution dates (INV-001 finding 5), and the runs were retained. The reason recorded is the scheduling of the laboratory system. The univariate study is judged on its own runs and on the set-point runs of the qualification study, and no result of the response-surface design depends on it, so the order changes no conclusion of this report.

13.3.6 PD-005-10. Pool host cell protein model used although its conclusion changes across the offset interval

PCMP-001 §7.1 states that a model whose conclusion for an output changes between the two ends of the offset interval is not used for that output. Over the NOR in force the upper prediction limit of a single run of pool host cell protein is 15,697 ng/mg with the offset ratio at its estimate and 17,143 ng/mg with the ratio at the low end of its interval, against the criterion of ≤ 16,719 ng/mg (Section 3.1). The conclusion therefore changes across the interval. The model is used all the same for the judgement over the NOR, for the PARs and for the design space, with the ratio at its estimate as PCMP-001 §7.1 rule 2 requires. The offset is known precisely enough to be accounted for at its estimate. The NOR comes from production and is not narrowed by an offset. The limitation is stated with each conclusion for pool host cell protein, and the protein load NOR is flagged for discussion with manufacturing, so that manufacturing can decide whether the NOR can be improved.

Table 13.4: Discrepancies of this report from its plan PCP-005, PD-005-01 to PD-005-04.

Discrepancy	Subject	Planned	Done	Reason
PD-005-01	Supplementary set-point runs in the scale-down model qualification	PCP-005: set-point runs of the scale-down model in study S-PA-SDM	further set-point runs in study S-PA-SDM2, made during the Version 2.0 re-analysis	to narrow the confidence interval of the offset for pool host cell protein, the output that sets the ranges of the step
PD-005-02	Conclusions resting on models that do not meet the criterion on residuals and influential runs	PCP-005: response-surface models meeting every model-adequacy criterion of the plan (criterion 5), used for the ranges and the design space, and each run with a Cook's distance above 4/n examined by deleting it and repeating the judgement	the models of the outputs that do not meet the criterion on residual patterns and influential runs (a residual trend at the level of the residual tests, or a run with a Cook's distance above the median of F(p, n − p)) used for the ranges and the design space, with the size of each trend stated against the criterion of its output and each run above the limit deleted in turn, the model selection repeated and the output judged over the NOR again	selected models of this step do not meet the criterion of PCMP-001 §7.2 on residual patterns and influential runs; whether a conclusion depends on the trend or on the run is shown by the size of the trend and by the deletion
PD-005-03	Response-surface run 23 analysed at its recorded elution buffer pH	PCP-005: every run analysed at its design setting	run 23 analysed at the elution buffer pH recorded under DEV-005-01, the setting the run was made at	the record of study event DEV-005-01 shows the buffer below its target pH; analysing the run at the recorded setting is the conservative reading
PD-005-04	Model selection for the response-surface outputs	PCP-005: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report

Table 13.5: Discrepancies of this report from its plan PCP-005, PD-005-05 to PD-005-07.

Discrepancy	Subject	Planned	Done	Reason
PD-005-05	Hold study of the Protein A eluate, made during the Version 2.0 re-analysis	PCP-005: no hold study; Version 1.0 of the plan did not foresee one	the Protein A eluate held at two temperatures for seven days, aggregate by SEC, acidic variants by icIEF and fragments by nrCE-SDS at every time point (study S-PA-HLD); the pool is judged against the criteria it must meet on entering the next step (PCMP-001 Version 2.0 §5), and the level expected to enter the hold and the reading of the change over it are the report's approach, chosen after the results were known	Version 2.0 judges a held intermediate against its intermediate acceptance criteria, and Version 1.0 had no hold result to judge
PD-005-06	Loaded antibody mass from the titre assay	PCP-005: step yield from the antibody mass in the load and in the pool, both by absorbance at 280 nm	the load mass from the product titre by Protein A affinity HPLC (SOP-3016), the pool mass by absorbance at 280 nm	absorbance at 280 nm measures every protein in a clarified harvest, not the antibody alone; the titre assay measures the antibody
PD-005-07	Univariate study without runs at the centre	PCP-005: each univariate parameter run at the ends of its range and at its centre	each end of each range run in duplicate, with no run at the centre; the qualification runs at set-point serve as the reference	the set-point reference was taken from the qualification runs

Table 13.6: Discrepancies of this report from its plan PCP-005, PD-005-08 to PD-005-10.

Discrepancy	Subject	Planned	Done	Reason
PD-005-08	Leached Protein A offset of the scale-down model accounted for, not investigated	PCP-005: Version 1.0: no statistically significant difference from the at-scale values at $p < 0.05$, and any difference investigated and the model modified before the characterization studies	leached Protein A differed between the scales at that level; it was not investigated before the characterization studies, and it is judged through its offset under PCMP-001 Version 2.0 §7.1	a scale-down model may differ in mean from manufacturing scale where its offset is known precisely enough, the rule of PCMP-001 Version 2.0 §7.1, so the Version 1.0 test was stricter than the use of the model needs
PD-005-09	Univariate study made before the response-surface design	PCP-005: the univariate assessments executed fourth, in parallel with or after the response-surface design (PCP-005 §11)	the univariate study S-PA-UNI ran after the screening design and before the response-surface design began	scheduling of the laboratory system
PD-005-10	Pool host cell protein model used although its conclusion changes across the offset interval	PCP-005: a model whose conclusion for an output changes between the two ends of the offset interval is not used for that output (PCMP-001 Version 2.0 §7.1)	the model of pool host cell protein is used for the judgement over the NOR, the PARs and the design space, judged with the offset ratio at its estimate (PCMP-001 Version 2.0 §7.1 rule 2); over the NOR in force the conclusion holds at the estimate and fails at the low end of the interval; the limitation is stated with each conclusion, and the protein load NOR is flagged for discussion with manufacturing	the offset is known precisely enough to be accounted for at its estimate, and the NOR comes from production and is not narrowed by an offset; the flag lets manufacturing decide whether the NOR can be improved

Appendices

Appendix A. Screening design matrix and responses

Table A.1 gives the screening design (study S-PA-SCR) in natural units, with the results as measured in the scale-down model. Table A.2 gives the run order of execution and the results measured for information.

Table A.1: Protein A screening design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW
		A g/L resin	B pH	C cm/hr	D CV	HCP ng/mg	Yield %	LPA ppm	SEC %
1		10	3.2	100	2	9,966	91.8	2.64	1.41
2		10	3.2	100	3.2	9,515	99.0	3.00	1.46
3		10	3.2	300	2	9,023	92.8	3.03	1.42
4		10	3.2	300	3.2	7,894	98.0	2.98	1.47
5		10	3.9	100	2	3,212	91.5	2.88	1.03
6		10	3.9	100	3.2	2,523	98.6	3.82	1.03
7		10	3.9	300	2	2,540	92.9	3.30	0.91
8		10	3.9	300	3.2	3,108	96.0	2.43	0.97
9	factorial	50	3.2	100	2	16,568	91.3	2.47	1.76
10		50	3.2	100	3.2	19,738	99.0	2.66	1.97
11		50	3.2	300	2	18,271	85.7	2.67	1.87
12		50	3.2	300	3.2	21,245	93.6	3.54	1.98
13		50	3.9	100	2	6,792	91.1	2.77	1.23
14		50	3.9	100	3.2	7,875	97.6	2.29	1.22
15		50	3.9	300	2	7,655	89.1	2.88	1.21
16		50	3.9	300	3.2	7,792	92.7	3.14	1.23
17		30	3.55	200	2.6	6,436	93.1	2.91	1.23
18	center	30	3.55	200	2.6	9,652	93.3	3.07	1.36
19		30	3.55	200	2.6	8,222	94.2	3.86	1.30

Factors: A, protein load. B, elution buffer pH. C, load flow rate. D, end of pool collect.
LPA, leached Protein A.
LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001).
Not tested in this design: LMW by nrCE-SDS.

Table A.2: Screening design (study S-PA-SCR), run order and results measured for information.

Run	Run order	HMW by SEC (%)	DNA, load (pg/mg)	DNA, pool (pg/mg)
1	16	1.65	n.t.	n.t.
2	19	1.59	n.t.	n.t.
3	5	1.55	n.t.	n.t.
4	13	1.57	n.t.	n.t.
5	6	1.50	n.t.	n.t.
6	18	1.53	n.t.	n.t.
7	12	1.40	n.t.	n.t.
8	3	1.50	n.t.	n.t.
9	7	1.59	n.t.	n.t.
10	15	1.66	n.t.	n.t.
11	2	1.59	n.t.	n.t.
12	10	1.66	n.t.	n.t.
13	1	1.48	n.t.	n.t.
14	8	1.54	n.t.	n.t.
15	14	1.52	n.t.	n.t.
16	9	1.47	n.t.	n.t.
17	17	1.49	220,796	1,372
18	4	1.47	183,600	1,068
19	11	1.51	184,942	941

Run order, the order of execution. HMW, aggregate by SEC. DNA, residual DNA by qPCR in pg per mg antibody, measured on the centre points only. n.t., not tested.

Appendix B. Response-surface design matrix and responses

Table B.1 gives the response-surface design (study S-PA-RSM) in natural units, with the results as measured in the scale-down model and the fragment results of study S-PA-RET. Table B.2 gives the run order of execution, which the trend test of criterion 5 uses, the elution buffer pH at which each run is analysed, and the results measured for information. Run 23 is analysed at its recorded pH (DEV-005-01).

Table B.1: Protein A response-surface design, settings and results.

Run	Type	Factor settings				Responses			Fragments, LMW	
		A	B	C	D	HCP	Yield	LPA	SEC	nrCE-SDS
		g/L resin	pH	cm/hr	CV	ng/mg	%	ppm	%	%
1	factorial	10	3.2	100	2	10,011	93.0	3.42	1.41	0.90
2		10	3.2	100	3.2	9,269	99.0	3.06	1.43	1.08
3		10	3.2	300	2	10,651	92.8	2.53	1.45	1.09
4		10	3.2	300	3.2	7,715	98.8	2.48	1.36	0.94
5		10	3.9	100	2	2,993	92.3	3.27	1.05	0.93
6		10	3.9	100	3.2	2,149	97.7	1.89	0.95	0.83
7		10	3.9	300	2	2,643	90.8	4.61	1.04	0.91
8		10	3.9	300	3.2	2,663	99.0	3.13	1.07	0.93
9		50	3.2	100	2	25,114	89.0	3.38	2.28	0.90
10		50	3.2	100	3.2	20,388	99.0	2.49	1.95	1.01
11[a)]		50	3.2	300	2	22,756	86.1	2.12	2.14	1.04
12		50	3.2	300	3.2	16,694	92.4	3.03	1.84	0.90
13		50	3.9	100	2	7,882	89.5	3.63	1.24	0.77
14		50	3.9	100	3.2	6,578	97.0	2.72	1.17	0.91
15		50	3.9	300	2	6,799	84.4	3.06	1.17	0.82
16		50	3.9	300	3.2	6,100	96.2	2.82	1.20	0.89
17	axial	10	3.55	200	2.6	3,900	93.5	2.71	1.14	1.00
18		50	3.55	200	2.6	9,377	93.5	3.37	1.36	0.87
19		30	3.2	200	2.6	13,732	96.6	3.77	1.64	0.92
20		30	3.9	200	2.6	4,122	95.5	3.80	1.08	0.85
21		30	3.55	100	2.6	7,571	95.5	2.30	1.32	0.87
22		30	3.55	300	2.6	8,098	95.5	2.54	1.29	0.98
23[b)]		30	3.55	200	2	8,167	90.8	2.07	1.36	1.00
24		30	3.55	200	3.2	6,461	95.3	2.69	1.22	0.96
25	center	30	3.55	200	2.6	8,036	94.0	2.75	1.39	0.97
26		30	3.55	200	2.6	7,617	96.7	1.70	1.29	0.85
27		30	3.55	200	2.6	7,206	92.4	4.06	1.24	0.93
28		30	3.55	200	2.6	7,396	95.4	2.91	1.37	0.80

[a)] DEV-005-02: Leached Protein A result of RSM run 11 initially out of trend; re-assayed. Run retained.
[b)] DEV-005-01: Elution-buffer pH prepared below target on an RSM run. Run retained, and analysed at the recorded elution buffer pH of 3.38 in Version 2.0.
Factors: A, protein load. B, elution buffer pH. C, load flow rate. D, end of pool collect.
LPA, leached Protein A.
LMW by SEC is biased in this matrix and is not used for evaluation (AMD-001).

Table B.2: Response-surface design (study S-PA-RSM), run order, elution buffer pH as analysed and results measured for information.

Run	Run order	Elution buffer pH, as analysed	HMW by SEC (%)	DNA, load (pg/mg)	DNA, pool (pg/mg)
1	18	3.2	1.68	n.t.	n.t.
2	20	3.2	1.68	n.t.	n.t.
3	12	3.2	1.59	n.t.	n.t.
4	8	3.2	1.73	n.t.	n.t.
5	1	3.9	1.43	n.t.	n.t.
6	14	3.9	1.51	n.t.	n.t.
7	28	3.9	1.43	n.t.	n.t.
8	16	3.9	1.42	n.t.	n.t.
9	27	3.2	1.64	n.t.	n.t.
10	26	3.2	1.60	n.t.	n.t.
11	10	3.2	1.62	n.t.	n.t.
12	23	3.2	1.58	n.t.	n.t.
13	24	3.9	1.45	n.t.	n.t.
14	4	3.9	1.51	n.t.	n.t.
15	5	3.9	1.52	n.t.	n.t.
16	15	3.9	1.50	n.t.	n.t.
17	19	3.55	1.59	n.t.	n.t.
18	13	3.55	1.54	n.t.	n.t.
19	7	3.2	1.62	n.t.	n.t.
20	22	3.9	1.58	n.t.	n.t.
21	11	3.55	1.63	n.t.	n.t.
22	21	3.55	1.55	n.t.	n.t.
23	3	3.38	1.51	n.t.	n.t.
24	2	3.55	1.52	n.t.	n.t.
25	25	3.55	1.54	226,078	966
26	9	3.55	1.53	234,778	827
27	17	3.55	1.55	137,045	635
28	6	3.55	1.58	150,553	498

Run order, the order of execution. HMW, aggregate by SEC. DNA, residual DNA by qPCR in pg per mg antibody, measured on the centre points only. n.t., not tested.

Appendix C. Full effect and coefficient tables

Table C.1 gives the screening effects on leached Protein A. The screening effects on pool host cell protein and step yield are in Table 5.4 and Table 5.5, and the coefficients of the selected response-surface models in Table 5.8, Table 5.11 and Table 5.12. The selected model of leached Protein A has no term besides its intercept.

Table C.1: Screening effects on leached Protein A (ppm), sorted by size.

Term	Effect	Coef.	Std. err.	t	p-value
AC	0.33	0.16	0.13	1.28	0.235
A	−0.21	−0.10	0.13	−0.81	0.441
BD	−0.19	−0.09	0.13	−0.74	0.482
BC	−0.18	−0.09	0.13	−0.71	0.497
C	0.18	0.09	0.13	0.70	0.503
D	0.15	0.08	0.13	0.59	0.571
AB	−0.13	−0.07	0.13	−0.52	0.618
CD	−0.10	−0.05	0.13	−0.39	0.710
B	0.07	0.03	0.13	0.25	0.806
AD	0.06	0.03	0.13	0.22	0.830

Appendix D. Analytical methods summary

Table D.1 summarizes the analytical methods used in the studies of the step.

Table D.1: Analytical methods used in the studies of the Protein A step.

Technique	Analyte	AMV	Reportable	Use in this report
ELISA	Host cell protein	AMV-3012	ng per mg antibody	Response of both designs, univariate study
	Leached Protein A	AMV-3016	ppm	Response of both designs, univariate study
Protein A affinity HPLC	Antibody in the load	AMV-3021	Antibody concentration	Protein load of each run, step yield
UV absorbance at 280 nm	Protein in the pool	AMV-3019	Protein concentration	Step yield
Non-reduced CE-SDS	Fragments (LMW)	AMV-3020	% of total	Response of the response-surface design, hold study
SEC-HPLC	Fragments (LMW)	AMV-3011	% of total	Information only
	Aggregate (HMW)	AMV-3011	% of total	Hold study, information
icIEF	Acidic variants	AMV-3013	% of total	Hold study
qPCR	Residual DNA	AMV-3014	pg per mg antibody	Information only

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